



AMRITA
VISHWA VIDYAPEETHAM

USER INTERFACE DESIGN LAB MANUAL

SUBMITTED BY:

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ROLL NO	AV.SC.U4CSE25136
YEAR/SEM	1 ST YEAR/SEM-2
CREDITS	3
SECTION	CSE-B

SUBMITTED TO:

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Week1:

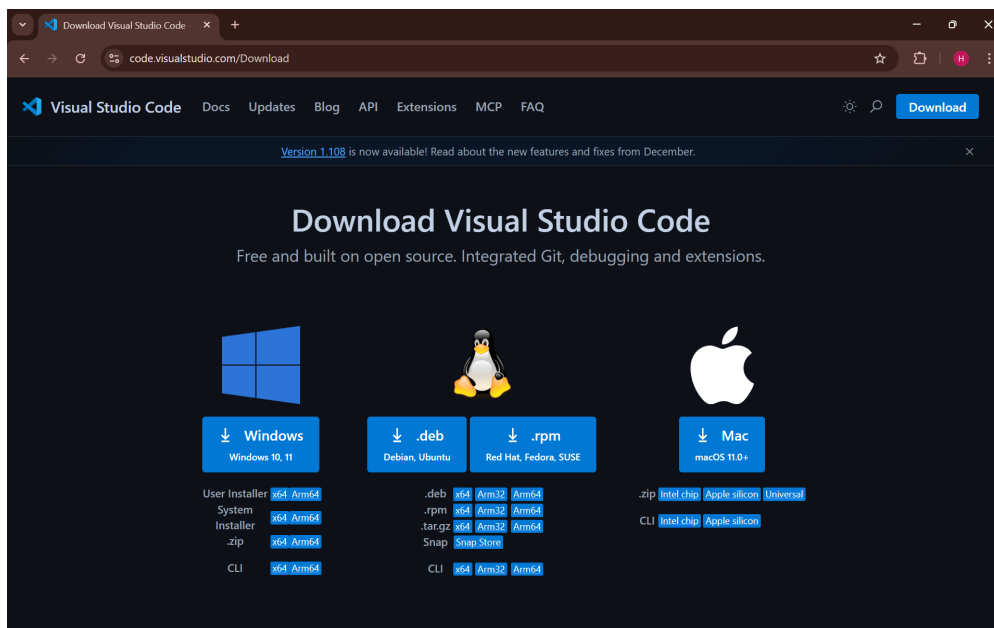
Lab programs:

1. Installation of VS Code tool.
2. Understanding basics of html stucture.
3. Understanding basic html tags.

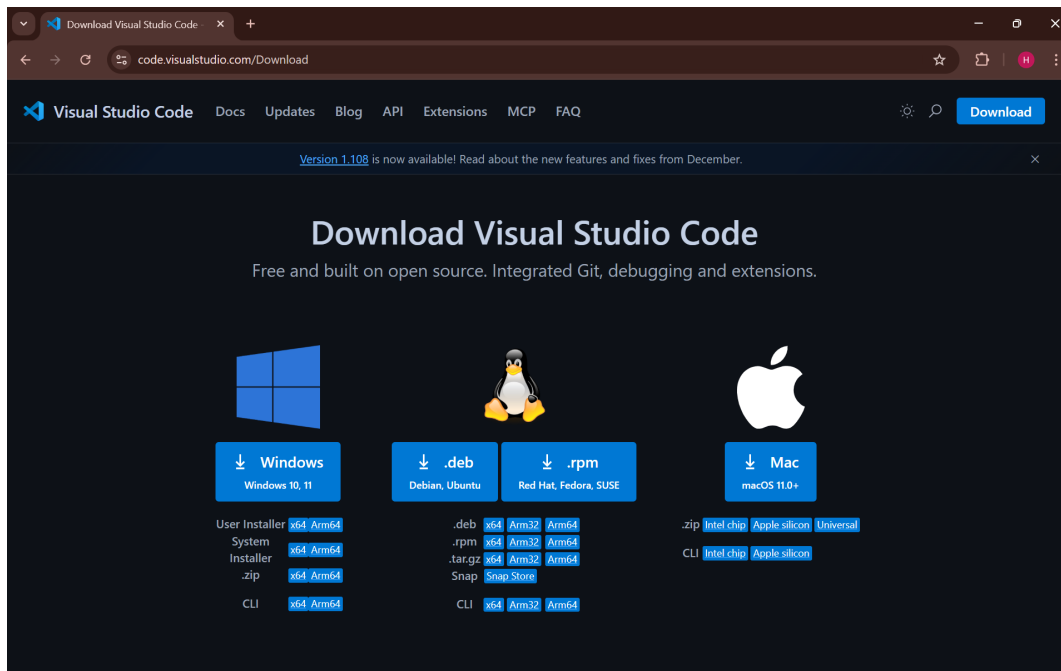
1.A) AIM: Downloading & Installation of VS code tool

Procedure:

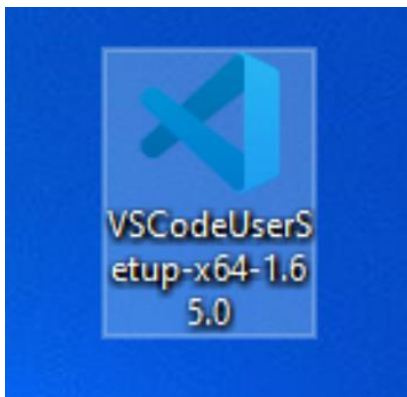
Step 1: Visit the Official Website of the Visual Studio Code using any web browser like [Google Chrome](#), [Microsoft Edge](#), etc.



[Step 2]: Press the "Download for Windows" button on the website to start the download of the Visual Studio Code Application.

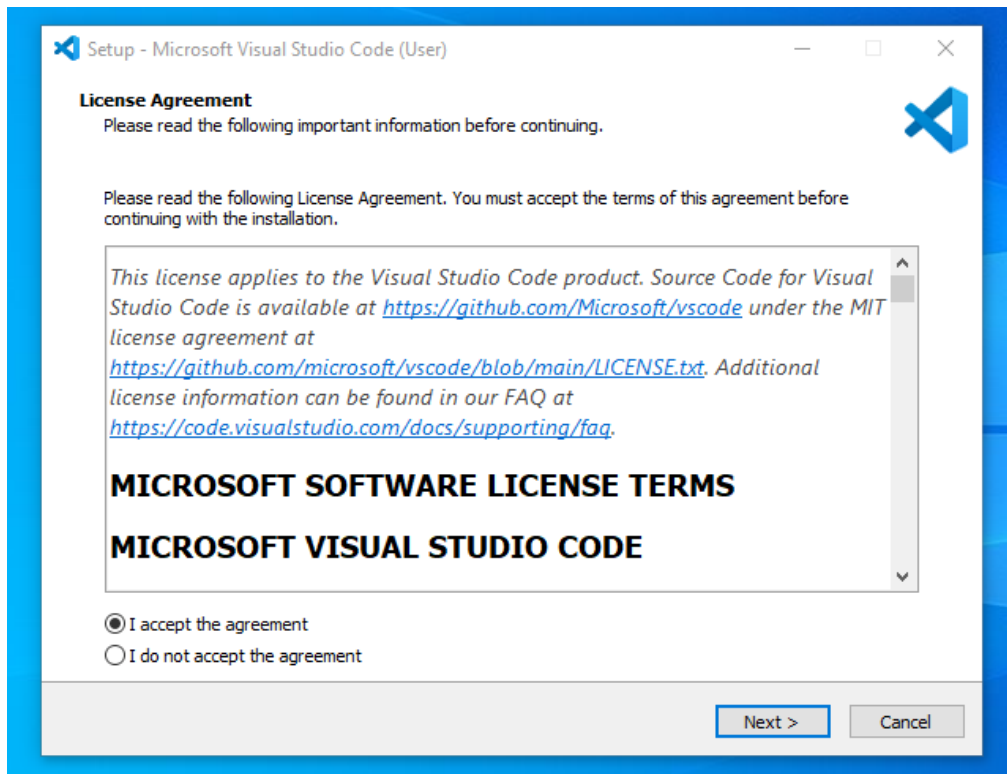


[Step3] When the download finishes, then the Visual Studio Code Icon appears in the downloads folder.

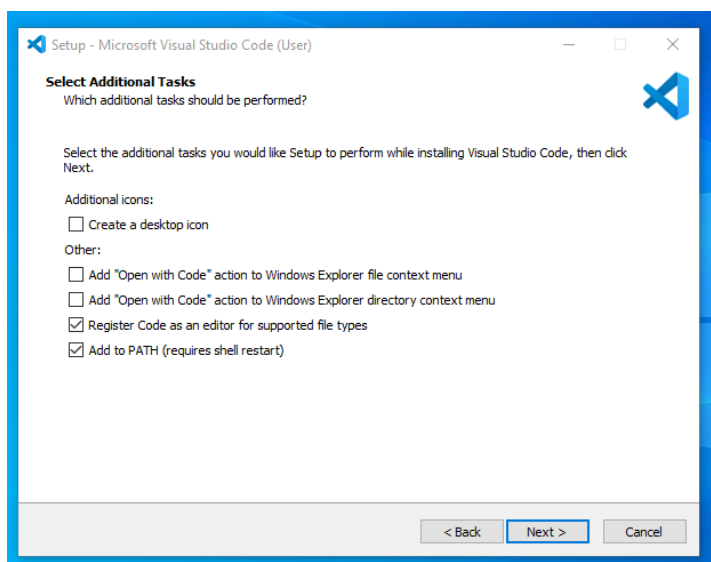


[Step4] Click on the Installer icon to start the installation process of the Visual Studio Code.

[Step5] After the Installer opens, it will ask you to accept the terms and conditions of the Visual Studio Code. Click on I accept the agreement and then click the Next button.

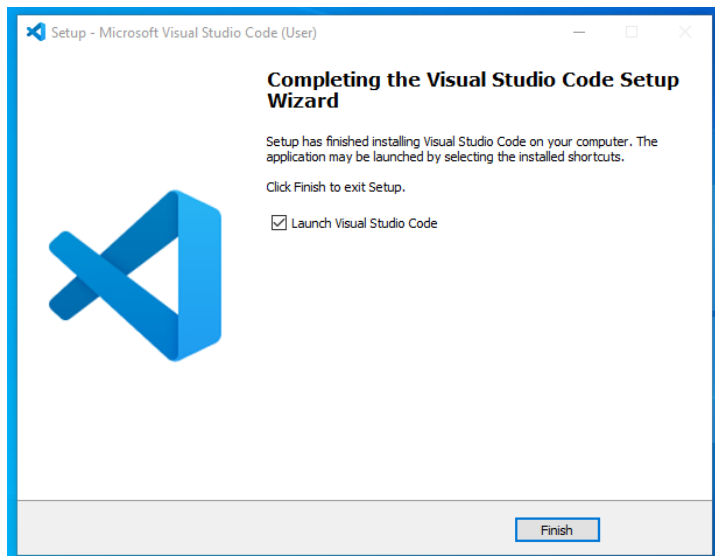


[Step6]:Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the Next button.

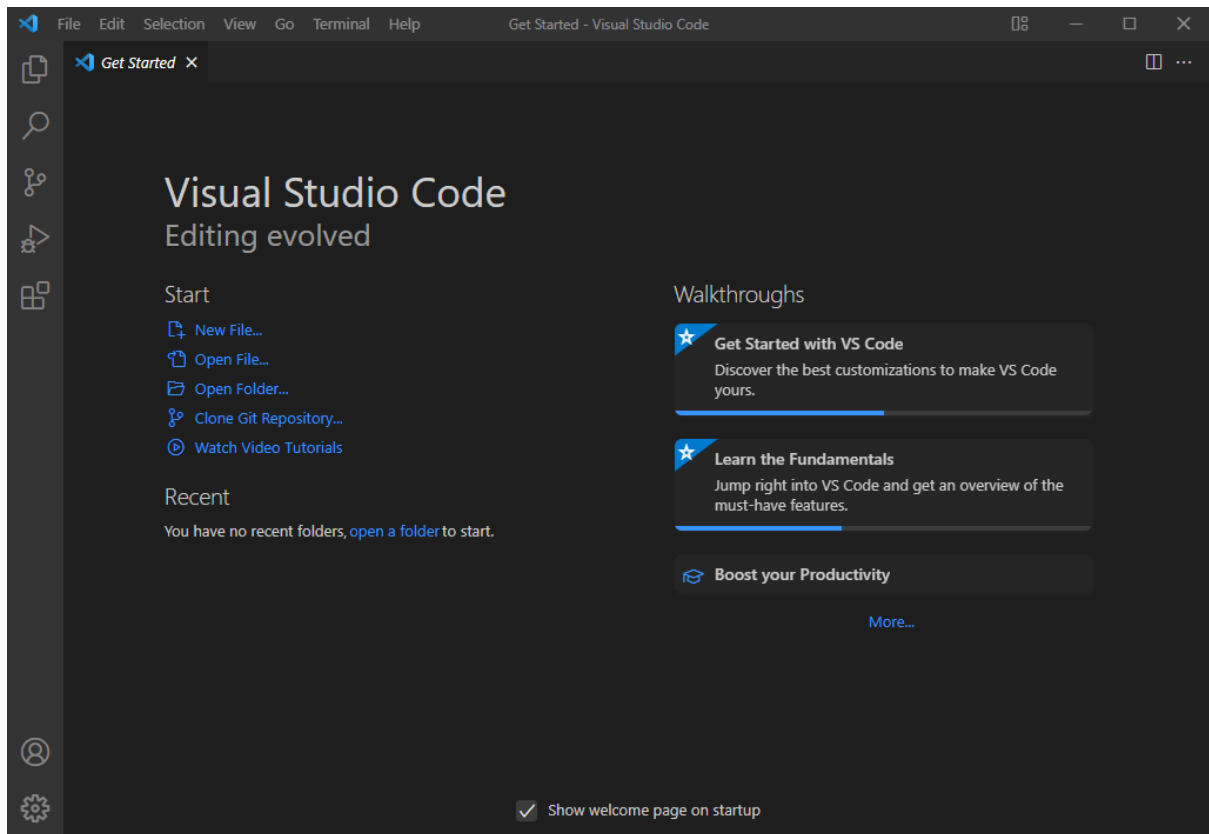


[Step7]Then it will ask to begin the installation setup. Click on the Install button. After clicking on Install, it will take about 1 minute to install the Visual Studio Code on your device.

After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the "Launch Visual Studio Code" checkbox and then click Next.



[Step 10]: After the previous step, the Visual Studio Code window opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!



So this is how we successfully installed Visual Studio Code on our Windows system. The steps mentioned in the above guideline can be used in any kind of Windows Browsers for Downloading & Installation VS Code on Windows 10. So, try these steps to get the VS Code IDE on your device as well.

1.B).Aim: To Understand the Structure of HTML

Procedure:

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```
<!DOCTYPE html>
```

```
<html>
```

```
<title>
```

```
    HTML BASICS
```

```
</title>
```

```
<body>
```

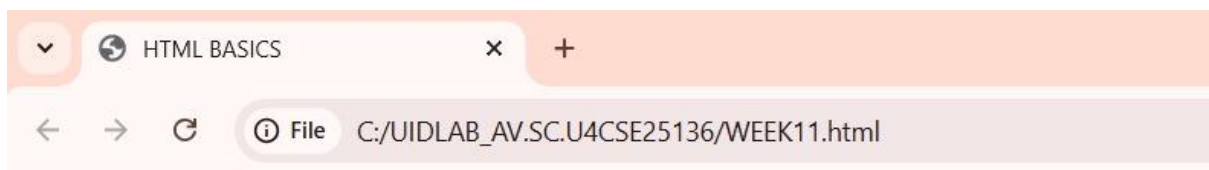
```
    <h1>FIRST HEADING</h1>
```

```
    <p>FIRST PARAGRAPH</p>
```

```
</body>
```

```
</html>
```

Output:



rithvik ;AV.SC.U4CSE25136 ;CSE B

FIRST HEADING

FIRST PARAGRAPH

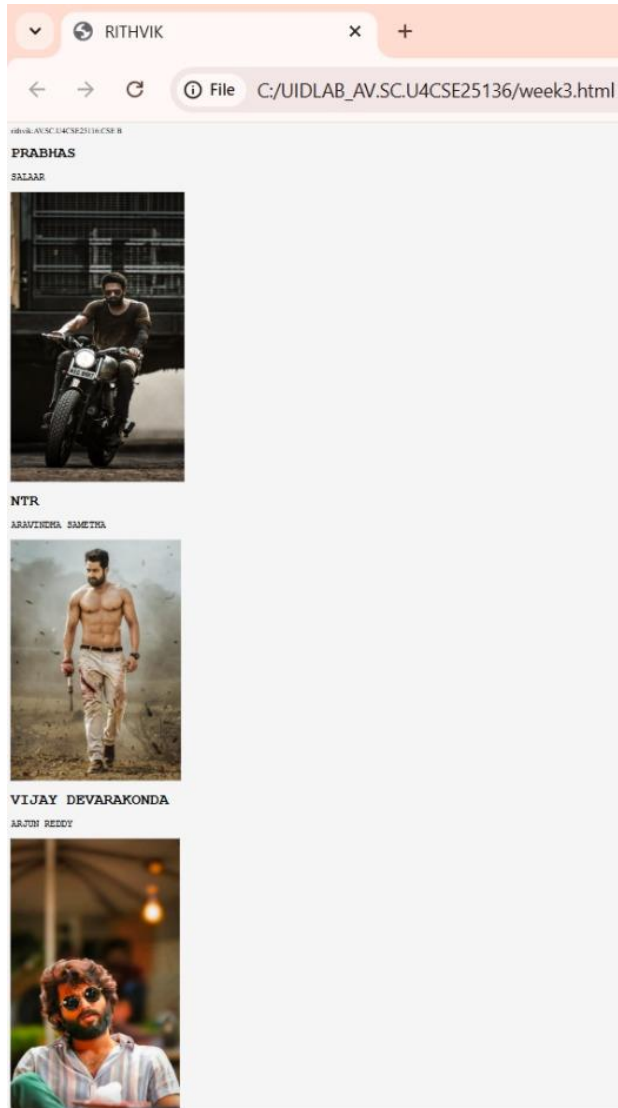
EXPLANATION:

- HTML: HYPER TEXT MARKUP LANGUAGE
- Used To Create And Structure Webpages
- `<!DOCTYPE HTML>`: tells that the doc type is html5
- `<html>`: The element is the root element of an HTML page.
- `<head>`: The element contains meta information about the HTML page
- `<title>`: The `<title>` element specifies a title for the HTML page.
- `<body>`: The `<body>` element contains all visible content like text images, link, tables, etc.
- `<h1>`: the `<h1>` element defines a large heading
- `<p>`: The `<p>` element defines a paragraph

1.C) Aim: To understand basic html tags

```
1 rithvik:AV.SC.U4CSE25116:CSE B
2 <!DOCTYPE HTML>
3 <html>
4
5     <head>
6     |   <title>RITHVIK</title>
7     | </head>
8     <body style="background-color: whitesmoke;">
9     |   <h1 style="font-family:Courier;font-size: xx-large;">PRABHAS</h1>
10    |   <p style="font-family: Courier;font-size: larger;">SALAAR</p>
11
12    |   <a href="https://www.instagram.com/actorprabhas/">
13    |   
14    |   </a>
15
16    |   <h1 style="font-family:Courier;font-size: xx-large;">NTR</h1>
17    |   <p style="font-family: Courier;font-size: larger;">ARAVINDHA SAMETHA</p>
18    |   <a href="https://www.instagram.com/jrntr/">
19    |   |   
20    |   |   </img>
21    |   </a>
22
23    |   <h1 style="font-family:Courier;font-size: xx-large;">VIJAY DEVARAKONDA</h1>
24    |   <p style="font-family: Courier;font-size: larger;">ARJUN REDDY</p>
25    |   <a href="https://www.instagram.com/theverakonda/">
26    |   |   
27    |   |   </img>
28    |   </a>
29
30    | </body>
31 </html>
```

Output:



Explanation:

- `<!DOCTYPE HTML>`: tells that the doc type is html5
- `<html>`: The element is the root element of an HTML page.
- `<head>`: The element contains meta information about the HTML page
- `<title>`: The `<title>` element specifies a title for the HTML page.
- `<body>`: The `<body>` element contains all visible content like text images, link, tables, etc.
- `<h1>`: the `<h1>` element defines a large heading
- `<p>`: The `<p>` element defines a paragraph
- `<img src="">`: it is used to add image
- `<a href="">`: it is used to add link in the html website

WEEK-2

2(A) To design a simple web page using HTML that demonstrates the use of background color, background image, heading styling, and preformatted text.

Program:

```
<!DOCTYPE html>

<html>

<head>

<title>Simple Web Page</title>


<style>

body{

    background-color: rgb(1, 255, 166);

    background-image: url("rithvik.jpg");

    background-size: cover;

}

h1{

    color: rgb(31, 31, 31);

    text-align: center;

    font-family: Arial;

}

h2{
```

```
    color: darkred;
    text-align: center;
}

pre{
    color: rgb(21, 22, 23);

    font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida
Sans Unicode', Geneva, Verdana, sans-serif;

    background-color: white;

    padding: 10px;

    border: 2px solid black;

    width: 400px;

    margin: auto;

    background-color: transparent;
}

</style>

</head>

<body>

<h1>Welcome to My Web Page</h1>

<h2>INTRODUCTION</h2>

<pre>
```

Name : ORAM RITHVIK REDDY

Course : Web Development

College : Amrita Vishwa Vidyapeetham

Hello! My name is ORAM RITHVIK REDDY

I am a B.Tech student .

I am interested in learning programming.

Currently, I am improving my skills .

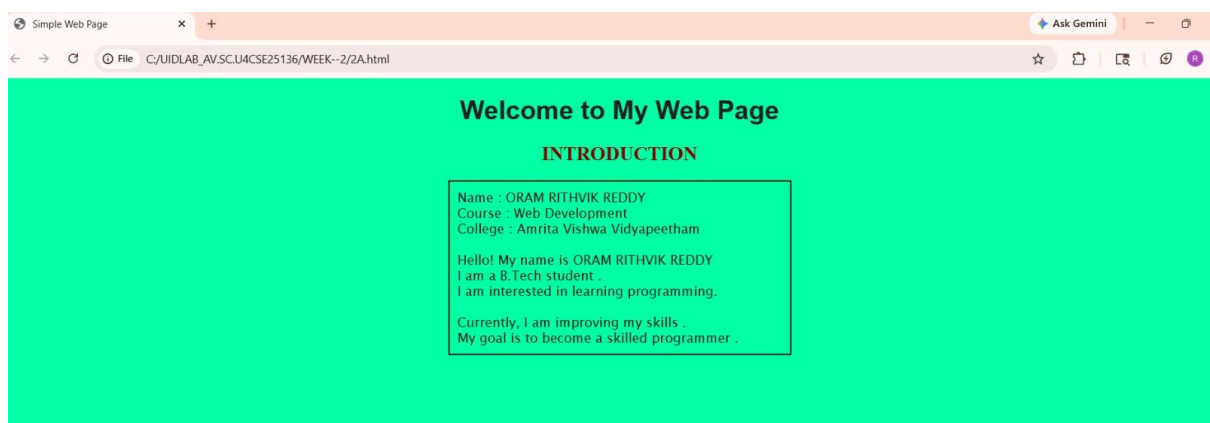
My goal is to become a skilled programmer .

</pre>

</body>

</html>

OUTPUT:



EXPLANATION:

Here is the explanation of the tags used in your HTML code, broken down into simple points:

Document Structure Tags

- **<!DOCTYPE html>**: Tells the browser that this is an **HTML5** document.
- **<html>**: The "root" or container for the entire web page. Everything else goes inside here.
- **<head>**: Contains **metadata** (information about the page) that isn't shown to the user, like titles and CSS styles.
- **<title>**: Defines the text that appears on the browser's **tab**.
- **<body>**: The container for all the **visible content** (text, images, headings) that people see on the screen.

Content & Text Tags

- **<h1>**: The primary heading. It is the largest and most important title on the page.
- **<h2>**: A secondary heading, used here for the "INTRODUCTION" section.
- **<pre>**: Stands for **Preformatted text**. Unlike regular tags, it displays text exactly as it is typed in the code—preserving every space, tab, and line break. It also uses a "monospaced" (fixed-width) font by default.

Styling Tags & CSS Properties

- **<style>**: Used to write **Internal CSS** to change the look of the page.
- **background-image: url(...)**: Sets an image as the backdrop of the page.
- **background-size: cover**: Stretches the background image to ensure it covers the entire screen without gaps.

- **text-align: center:** Moves the text to the middle of the page horizontally.
- **font-family:** Defines which font (like Arial or Verdana) the text should use.
- **margin: auto:** Centers the <pre> box horizontally on the page.
- **padding:** Adds space **inside** the box so the text doesn't touch the borders.

2(B) Design a basic webpage with heading, paragraph, pre, table by adding styles.

```
<!DOCTYPE html>
<html>
  <head>
    <title>website</title>
  </head>

  <body style="background-color:rgb(231, 4, 4);background-
image:url('amrita.jpg');">
    <h1 style="text-align:center;color:crimson;border: 7px solid rgb(238,
155, 130);">AMRITA VISHWA VIDYAPEETHAM</h1>

    <pre style="text-align: center;font-style:italic;color:rgb(255, 255,
255);font-size:x-large;">
      Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and
research-intensive

      university, establishing its campus in Amaravati. Nestled amidst
the scenic beauty of the

      Krishna River and surrounded by natural wonders and historical
sites, Amaravati city

      embodies the principles of sustainability and liveability. We are
building one of the most

      buoyant centre of the research, education and technological
advancement, in the all new

      greenfield capital city.

    <table border="1" cellpadding="5" cellspacing="10" width="60%"
align="center" bgcolor="beige" style="color:rgb(0, 0, 0);font-family: Georgia,
'Times New Roman', Times, serif;">
      <tr>
        <th>slab</th>
        <th>Amaravati</th>
        <th>Benguluru</th>
        <th>coimbature</th>
        <th>chennai</th>
      </tr>

      <tr>
        <td>1.</td>
        <td>1,25,000</td>
```

```

        <td colspan="2">1,50,000</td>
        <!-- <td>1,50,000</td> -->
        <td>1,75,000</td>
    </tr>

    <tr>
        <td rowspan="2">2.</td>
        <td>2,25,000</td>
        <td>2,50,000</td>
        <td>2,75,000</td>
        <td>2,40,000</td>
    </tr>

    <tr>
        <td>3.</td>
        <td colspan="2">3,25,000</td>
        <!-- <td>3,25,000</td> -->
        <td>3,50,000</td>
        <td>3,75,000</td>
    </tr>
</table>
</pre>
</body>
</html>

```

OUTPUT:

AMRITA VISHWA VIDYAPEETHAM

Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and research-intensive university, establishing its campus in Amaravati. Nestled amidst the scenic beauty of the Krishna River and surrounded by natural wonders and historical sites, Amaravati city embodies the principles of sustainability and liveability. We are building one of the most buoyant centres of the research, education and technological advancement, in the all new greenfield capital city.

slab	Amaravati	Benguluru	coimbatore	chennai
1.	1,25,000	1,50,000	1,75,000	1,75,000
2.	2,25,000	2,50,000	2,75,000	2,40,000
3.	3,25,000	3,50,000	3,75,000	3,75,000

EXPLANATION:

1. The `<!DOCTYPE html>` declaration defines that the document is an HTML5 webpage.
2. The `<html>` tag is the root element that contains the entire webpage content.
3. Inside the `<head>` section, the `<title>` tag sets the browser tab name as "website".
4. The `<body>` tag contains all the visible content of the webpage.
5. The body uses inline CSS to set a red background color and a background image (amrita.jpg).
6. The `<h1>` tag displays the heading "AMRITA VISHWA VIDYAPEETHAM", centered with a crimson color and a solid border.
7. The `<pre>` tag is used to preserve spaces and line breaks, keeping the paragraph formatting exactly as written.
8. Inside the `<pre>` tag, a descriptive paragraph explains about Amrita Vishwa Vidyapeetham and its Amaravati campus.
9. A `<table>` is created with border, padding, spacing, width, alignment, background color, and font styling using inline CSS.
10. The first row of the table contains header cells (`<th>`) representing different campus locations.
11. The `colspan="2"` attribute merges two columns into one cell in certain rows.
12. The `rowspan="2"` attribute merges two rows into a single cell vertically.
13. Some `<td>` elements are commented out using `<!-- -->`, so they will not appear in the output.

2(C) Aim:

To design a restaurants menu webpage using HTML and displaying menu items in a table format with proper layout and visual styling.

Program:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>MENU CARD</title>
```

```
<style>
```

```
    td{
```

```
        color:rgb(255, 0, 0);
```

```
    }
```

```
</style>
```

```
</head>
```

```
<body style="font-family: 'Times New Roman', Times, serif; background-size: cover; ">
```

```
<h1 style="text-align: center; border: 5px solid #c00; margin: 0; color: #fff; background-color:
```

```
rgba(30, 10, 10, 0.75);">
```

```
<u><strong>RESTAURANT MENU</strong></u>
```

```
</h1>
```

```
<table style="width: 100%; color:whitesmoke; text-align: center;">
```

```

<tr style="background-color: #a00;">
<th>S.No</th>
<th>Dish</th>
<th>PRICE (in ₹)</th>
<th>Add-ons</th>
</tr>

<tr>
<td>1</td>
<td>Paneer Tikka </td>
<td><del>300</del> <strong>250</strong></td>
<td>Mint Chutney</td>
</tr>

<tr style="background-color: rgba(255,255,255,0.05);">
<td>2</td>
<td>Chicken Biryani <i></i> <sub>Handi
Style</sub></td>
<td><del>400</del> <strong>350</strong></td>
<td>Extra Raita</td>
</tr>

<tr>
<td>3</td>
<td>Veg Burger <sub>Double Patty</sub></td>
<td><strong>120</strong></td>
<td>Cheese Slice</td>
</tr>

```

4	Chocolate Lava Cake ^{New} _{With Molten Center}
200 150	Vanilla Scoop
5	Masala Dosa _{Butter Roast}
90	Extra Sambar
6	Butter Chicken <i>[CHEF'S SPECIAL]</i>
450 380	Butter Naan
7	Dal Makhani _{Slow Cooked}
220	Lacha Paratha

<tr style="background-color: rgba(255,255,255,0.05);">

<td>8</td>

<td>Margherita Pizza _{Thin Crust}</td>

<td>350 299</td>

<td>Extra Oregano</td>

</tr>

<tr>

<td>9</td>

<td>Veg Manchurian _{Gravy Base}</td>

<td>180</td>

<td>Fried Rice</td>

</tr>

<tr style="background-color: rgba(255,255,255,0.05);">

<td>10</td>

<td>Pasta Alfredo <sub>White

Sauce</sub></td>

<td>320 260</td>

<td>Garlic Bread</td>

</tr>

<tr>

<td>11</td>

<td>Fish & Chips _{Crispy Batter}</td>

<td>420</td>

<td>Tartar Sauce</td>

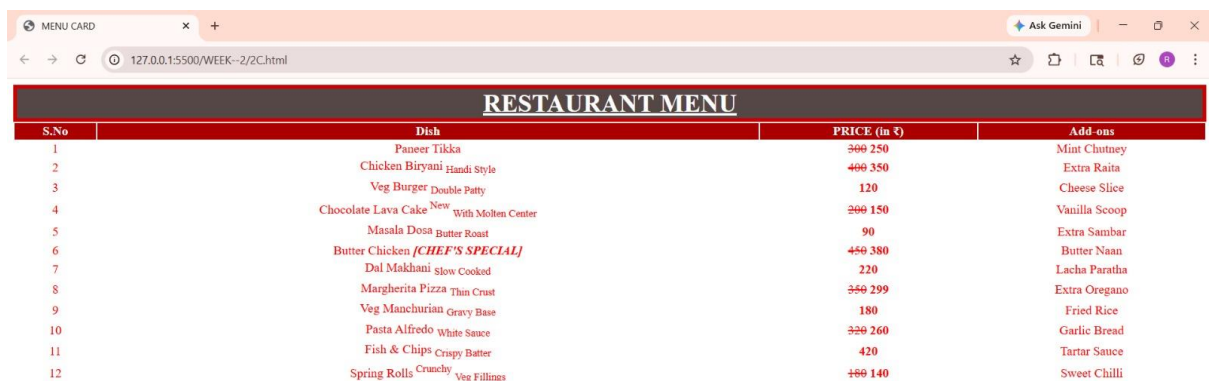
</tr>

```

<tr style="background-color: rgba(255,255,255,0.05);">
<td>12</td>
<td>Spring Rolls<sup> Crunchy</sup> <sub>Veg Fillings</sub></td>
<td><del>180</del> <strong>140</strong></td>
<td>Sweet Chilli</td>
</tr>
</table>
</body>
</html>

```

OUTPUT:



S.No	Dish	PRICE (in ₹)	Add-ons
1	Paneer Tikka	200 250	Mint Chutney
2	Chicken Biryani Handi Style	400 350	Extra Raita
3	Veg Burger Double Patty	120	Cheese Slice
4	Chocolate Lava Cake New With Molten Center	200 150	Vanilla Scoop
5	Masala Dosa Butter Roast	90	Extra Sambar
6	Butter Chicken [CHEF'S SPECIAL]	450 380	Butter Naan
7	Dal Makhani slow Cooked	220	Lacha Paratha
8	Margherita Pizza Thin Crust	350 299	Extra Oregano
9	Veg Manchurian Gravy Base	180	Fried Rice
10	Pasta Alfredo White Sauce	320 260	Garlic Bread
11	Fish & Chips Crispy Batter	420	Tartar Sauce
12	Spring Rolls Crunchy Veg Fillings	180 140	Sweet Chilli

EXPLANATION:

1. **<!DOCTYPE html>** – Declares the document as HTML5.
2. **<html>** – Root element that contains the entire webpage.
3. **<head>** – Contains metadata like title and styles.
4. **<title>** – Sets the title shown in the browser tab.
5. **<style>** – Internal CSS to style elements (here, td text color is red).
6. **style attribute** – Inline CSS used inside tags for styling (background, font, alignment).
7. **<body>** – Contains all visible content of the webpage.
8. **<h1>** – Main heading (“RESTAURANT MENU”).
9. **<u>** – Underlines the text.
10. **** – Makes text bold and important.
11. **<table>** – Creates the table for menu display.
12. **<tr>** – Defines a table row.
13. **<th>** – Table header cell (bold and centered).
14. **<td>** – Table data cell (normal content).
**** – Shows deleted (old) price with strike-through.
15. **<sub>** – Subscript text (dish descriptions).
16. **<sup>** – Superscript text (like “New”).
17. **<i>** – Italic text (for styling like chef’s special).

WEEK-3

3A. Construct a class time-table using html tags

```
<!DOCTYPE html>

<html>
<head>
  <title>TIMETABLE</title>
</head>
<body>
  <P>NAME : O RITHVIK REDDY SEC: CSE B ROLL NO: CSE25136</P>
  <table border="2" style="border-collapse: collapse;" width="100%"
align="center">
    <tr>
      <th>Time/Day</th>
      <th>SLOT 1</th>
      <th>SLOT 2</th>
      <th>10.35-10.45</th>
      <th>SLOT 3 </th>
      <th>SLOT 4 </th>
      <th>SLOT 5</th>
      <th>1.15-2.05</th>
      <th>SLOT 6</th>
      <th>3.00-3.10</th>
      <th>SLOT 7</th>
      <th>SLOT 8</th>
    </tr>
    <tr>
      <td>MONDAY</td>
      <td colspan="2">LINEAR ALGEBRA LAB</td>
      <td class="" rowspan="5" style="background-color: rgb(238, 255,
0);">short break</td>
      <td>GLIMPSE OF GLORIOUS</td>
      <td>MODERN PHYSICS</td>
      <td>LIBRARY</td>
      <td rowspan="5" style="background-color: rgb(238, 255, 0);">Lunch
Break</td>
      <td>MANUFACTURING LAB</td>
      <td rowspan="5" style="background-color: rgb(238, 225, 0);">Short
break</td>
      <td colspan="2">MANUFACTURING LAB</td>
    </tr>
    <tr>
      <td>TUESDAY</td>
      <td>DISCRETE MATHMATICS</td>
      <td>LINEAR ALGEBRA</td>
      <td>MODREN PHYSICS</td>
```

```

        <TD>OBJECT ORIENTED PROGRAMMING</TD>
        <TD>USER INTERFERENCE DESIGN</TD>
        <TD>GLIMPSE OF GLORIOUS</TD>
        <TD colspan="2">LIBRARY/SPORTS</TD>
    </TR>
    <TR>
        <TD>WEDNESDAY</TD>
        <TD colspan="2">OBJECT ORIENTED PROGRAMMING LAB</TD>
        <TD>DISCRETE MATHEMATICS</TD>
        <TD>USER INTERFERENCE DESIGN</TD>
        <TD>GLIMPSE OF GLORIOUS</TD>
        <TD>LIBRARY/SPORTS</TD>
        <TD colspan="2">LIBRARY/SPORTS</TD>
    </TR>
    <TR>
        <TD>THURSDAY</TD>
        <TD>LINEAR ALGEBRA</TD>
        <TD>OBJECT ORIENTED PROGRAMMING</TD>
        <TD>DISCRETE MATHEMATICS</TD>
        <TD style="color: red;">COUNSELLING</TD>
        <TD>LIBRARY</TD>
        <TD>MODERN PHYSICS</TD>
        <TD colspan="2">LIBRARY/SPORTS</TD>
    </TR>
    <TR>
        <TD>FRIDAY</TD>
        <TD colspan="2">USER INTERFERENCE DESIGN LAB</TD>
        <TD>LINEAR ALGEBRA</TD>
        <TD>OBJECT ORIENTED PROGRAMMING</TD>
        <TD>LIBRARY/SPORTS</TD>
        <TD style="color: red;">COUNSELLING</TD>
        <TD colspan="2">DISCRETE MATHEMATICS LAB</TD>
    </TR>
</table>
<br>
<hr>
<br>

<table width="100%" border="1" cellpadding="1" cellspacing="2"
style="text-align: center;background-color: rgb(255, 255, 255);">
    <tr>
        <th>Course code</th>
        <th>Course Name</th>
        <th> L-T-P</th>
        <th> Credits</th>
        <th>Faculty Name</th>
        <th> Assisting Faculty</th>
    </tr>

```



```

<td>23MAT116</td>
<td>Discrete Mathematics</td>
<td> 3-0-2 </td>
<td>04</td>
<td> Dr.Srivivasa Rao Thota</td>
<td> </td>
<tr>
  <td>23MAT117</td>
  <td>Linear Algebra</td>
  <td> 3-0-2 </td>
  <td>04</td>
  <td> Mr.Sapavat Bixapathi</td>
  <td> </td>
</tr>
<tr>
  <td>23CSE111</td>
  <td>Object Oriented Programming</td>
  <td> 3-0-4</td>
  <td>04</td>
  <td>Dr. Balaji TK</td>
  <td> </td>
</tr>
<tr>
  <td>23PHY115</td>
  <td>Modern Physics</td>
  <td> 2-1-0</td>
  <td>03</td>
  <td>Dr. C R Kesavalu</td>
  <td> </td>
</tr>
<tr>
  <td>23CSE113</td>
  <td>User Interface Design</td>
  <td> 2-0-2 </td>
  <td>03</td>
  <td> Dr.Rajkumar Batchu</td>
  <td> Mr. Barge Bharadwaj</td>
</tr>
<tr>
  <td>23MEE115</td>
  <td>Manufacturing Practice</td>
  <td> 0-0-3</td>
  <td>01</td>
  <td>Dr. Sathies S.</td>
  <td> </td>
</tr>
<tr>
  <td>22ADM111</td>

```

```

<td>Glimpses of Glorious India</td>
<td> 2-0-1</td>
<td>02</td>
<td>Dr. Siva Panuganti</td>
<td> </td>
</tr>
<tr>
<td> </td>
<td>Total (15L + 1Tut +12P =28 hrs)</td>
<td>15-1-12</td>
<td>21</td>
<td> </td>
<td> </td>
</tr>
</body>
</html>

```

OUTPUT:

NAME : O RITHVIK REDDY SEC: CSE B ROLL NO: CSE25136											
Time/Day	SLOT 1	SLOT 2	10.35-10.45	SLOT 3	SLOT 4	SLOT 5	1.15-2.05	SLOT 6	3.00-3.10	SLOT 7	SLOT 8
MONDAY	LINEAR ALGEBRA LAB			GLIMPSE OF GLORIOUS	MODERN PHYSICS	LIBRARY		MANUFACTURING LAB		MANUFACTURING LAB	
TUESDAY	DISCRETE MATHMATICS	LINEAR ALGEBRA		MODREN PHYSICS	OBJECT ORIENTED PROGRAMMING	USER INTERFERNCE DESIGN		GLIMPSE OF GLORIOUS		LIBRARY/SPORTS	
WEDNESDAY	OBJECT ORIENTED PROGRAMMING LAB		short break	DISCRETE MATHMATICS	USER INTERERENCE DESIGN	GLIMPSE OF GLORIOUS	Lunch Break	LIBRARY/SPORTS	Short break	LIBRARY/SPORTS	
THURSDAY	LINEAR ALGEBRA	OBJECT ORIENTED PROGRAMING		DISCRETE MATHMATICS	COUNSELLING	LIBRARY		MODREN PHYSICS		LIBRARY/SPORTS	
FRIDAY	USER INTERFERENCE DESIGN LAB			LINEAR ALGEBRA	OBJECT ORIENTED PROGRAMMING	LIBRARY/SPORTS		COUNSELLING		DISCRETE MATHMATICS LAB	

Course code	Course Name	L-T-P	Credits	Faculty Name	Assisting Faculty
23MAT116	Discrete Mathematics	3-0-2	04	Dr.Srivivasa Rao Thota	
23MAT117	Linear Algebra	3-0-2	04	Mr.Sapavat Bixapathi	
23CSE111	Object Oriented Programming	3-0-4	04	Dr. Balaji TK	
23PHY115	Modern Physics	2-1-0	03	Dr. C R Kesavalu	
23CSE113	User Interface Design	2-0-2	03	Dr.Rajkumar Batchu	Mr. Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	01	Dr. Sathies S.	
22ADM111	Glimpses of Glorious India	2-0-1	02	Dr. Siva Panuganti	
	Total (15L + 1Tut +12P =28 hrs)	15-1-12	21		

Explanation:

1. `<!DOCTYPE html>` defines the document type and tells the browser that this is an HTML5 document.
2. The `<html>` tag is the root element that contains the entire webpage.
3. Inside `<head>`, the `<title>` tag sets the page title as “TIMETABLE”, which appears in the browser tab.
4. The `<body>` section contains all visible content of the webpage.
5. A `<p>` tag is used to display student details (Name, Section, Roll Number).
6. The first `<table>` creates the weekly timetable with `border="2"` and `border-collapse: collapse` for better styling.
7. `<tr>` defines table rows, `<th>` defines header cells, and `<td>` defines data cells.
8. `colspan` is used to merge multiple columns (e.g., lab sessions occupying two slots).
9. `rowspan` is used to merge rows vertically (e.g., Short Break and Lunch Break across multiple days).
10. Inline CSS (`style`) is used to add background colors and text color (e.g., yellow for breaks, red for counselling).
11. The second `<table>` displays course details like course code, name, L-T-P structure, credits, and faculty names.
12. Attributes like `cellpadding`, `cellspacing`, `width`, and `alignment` improve spacing and presentation of the table.

3B. Construct a web page that should display that syllabus of each subject in image format also display the current semester time-table.

```
<!DOCTYPE html>
<html>
<head>
  <title>Document</title>
</head>
<body>

  <!DOCTYPE html>
<html>
<head>
  <title>TIMETABLE</title>
</head>
<body>
  <P>NAME : O RITHVIK REDDY SEC: CSE B ROLL NO: CSE25136</P>

  <h1 style="text-align: center;color: rgb(255, 64, 64);">Syllabus</h1>

  <h2 style="color: blueviolet;">Discrete Mathematics</h2>
  

  <h2 style="color: blueviolet;">Linear Algebra</h2>
  

  <h2 style="color: blueviolet;">Object Oriented Programming</h2>
  

  <h2 style="color: blueviolet;">Modern Physics</h2>
  

  <h2 style="color: blueviolet;">User Interface Design</h2>
  

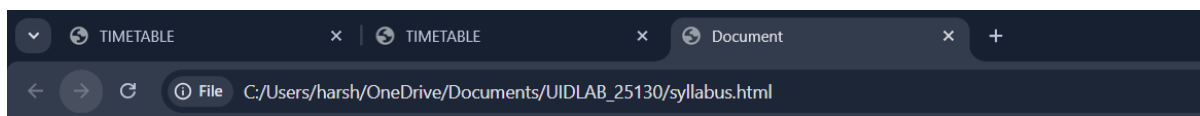
  <h2 style="color: blueviolet;">Manufacturing Practice</h2>
  

  <h2 style="color: blueviolet;">Glimpses of Glorious India</h2>
```

```


</body>
</html>
```

OUTPUT:



Syllabus

Discrete Mathematics

Syllabus

Unit 1

Logic, Mathematical Reasoning and Counting: Logic, Propositional Equivalence, Predicate and Quantifiers, Theorem Proving, Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2

Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.

Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3

Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem. Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

Lab Practice Problems: Verifications of logical statements, truth table, tautology. Recursive algorithms. Graph problems, degree, shortest path algorithm, Euler's algorithm and closest neighbour algorithm.

Textbook(s)

Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw- Hill Publishing Company Limited, New Delhi, Sixth Edition, 2007.
James Strayer, "Elementary Number Theory", Waveland Press, 2002.

Linear Algebra

Syllabus

Unit 1

Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis - Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2

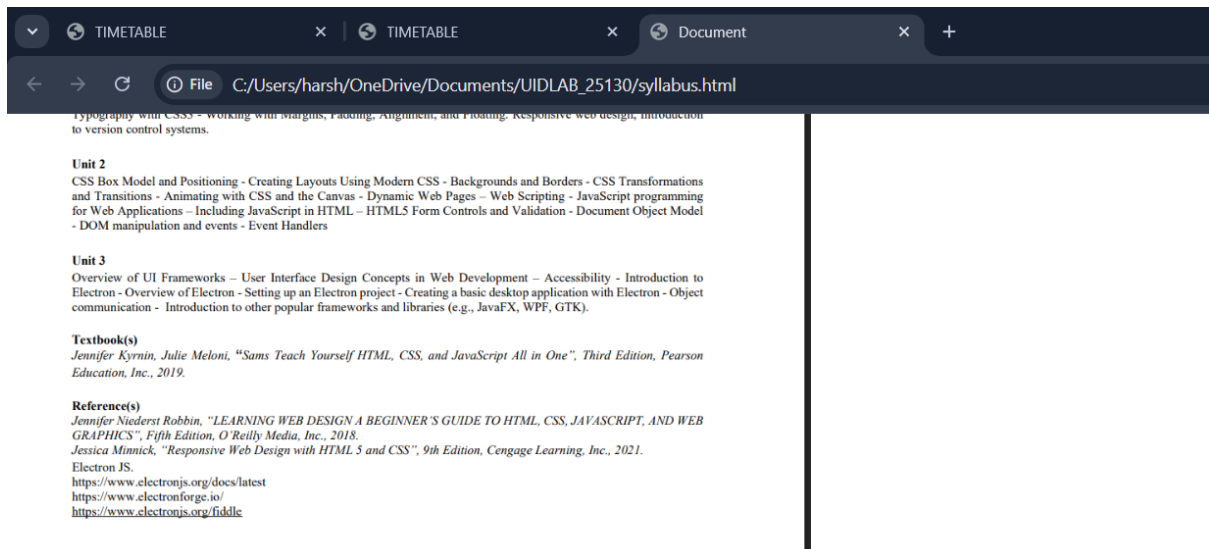
Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations. Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3

Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

Textbook(s)

Howard Anton and Chris Rorres, "Elementary Linear Algebra", Tenth Edition, John Wiley & Sons, 2010.



Manufacturing Practice

Syllabus

1. Additive Manufacturing Laboratory –12 hours

Introduction to digital manufacturing. Introduction to Additive Manufacturing - types - additive manufacturing applications - Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation- G code generation - 3D printing of simple geometries.

2. Mechanical Engineering Laboratory –12 hours

Study of tools and equipment used for basic manufacturing processes.

Manual arc welding practice for making Butt and Lap joints - Soldering Practice

Introduction to Machine Tools and Machining Processes

3. Automation lab –12 hours

Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.

4. Automobile Engineering lab –9 hours

Overview of automobiles – components –functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

Reference(s)

Glimpses of Glorious India

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

Explanation:

This program is an HTML webpage that shows the 2nd semester syllabus. It starts with the basic HTML structure and sets the ppage background color, and a centered heading “SYLLABUS” is displayed at the top.

Next, the program lists each subject name using <h2> headings and shows the syllabus images using tags.

WEEK-4

4A. A College wants to collect student details through an online admission form. Design a registration form using HTML tags (no CSS) with proper labels, table alignment, background color, and borders.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>student form</title>
</head>
<style>
  body{
    background-color: rgb(0, 255, 195);
  }
  table{
    background-color:rgb(220, 255, 195);
  }
  td{
    padding: 5%;
    column-width : 1000px;;
  }
</style>
<body>
  <form>
    <h2 style="text-align: center;color:rgb(0, 0, 0);">COLLEGE
ADMINISTRATION REGISTRATION</h2>

    <table border="3" align="center" style="height: 1000px; width:
550px;">
      <tr>
        <td>
          <label for="fname">FULL NAME</label>
        </td>
        <td>
          <input type="text" id="fname" placeholder="Fullname">
        </td>
      </tr>

      <tr>
        <td>
          <label for="email">E-mail</label>
        </td>
        <td>
```

```

        <input type="email" id="email"
placeholder="abcd@gmail.com">
    </td>
</tr>

<tr>
    <td>
        <label for="phone">Phone Number</label>
    </td>
    <td>
        <input type="number" id="phone" placeholder="Phone
Number">
    </td>
</tr>

<tr>
    <td>
        <label for="gender">Gender</label>
    </td>
    <td>
        <input type="radio" id="male" name="gender">
        <label for="male">Male</label>
        <input type="radio" id="Female" name="gender">
        <label for="Female">Female</label>
        <input type="radio" id="Other" name="gender">
        <label for="Other">Other</label>
    </td>
</tr>

<tr>
    <td> <label for="course">Course Applying for</label></td>

    <td>
        <select id="course">
            <option value="">--Select--</option>
            <option value="B.Tech">B.Tech</option>
            <option value="M.Tech">M.Tech</option>
            <option value="M.B.A">M.B.A</option>
            <option value="Other">Other</option>
        </select>
    </td>
</tr>

<tr>
    <td> <label for="hobbies">Hobbies</label></td>

    <td>

```



```

        <input type="checkbox" id="Sports">
        <label for="Sports">Sports</label>

        <input type="checkbox" id="Music">
        <label for="Music">Music</label>

        <input type="checkbox" id="Reading">
        <label for="Reading">Reading</label>

    </td>
</tr>

<tr>
    <td><label for="date">Date</label></td>

    <td> <input type="date" id="date"> </td>
</tr>

<tr>
    <td><label for="certificates">Upload Certificates</label></td>
    <td>
        <input type="file" id="certificates" multiple>
    </td>
</tr>

<tr>
    <td><label for="address">Address</label></td>
    <td>
        <textarea rows="7" cols="30" placeholder="Enter your
address here!!"></textarea>
    </td>
</tr>

<tr>
    <td colspan="2" style="padding-left: 140px;">
        <button type="reset" onclick="alert('Reset
Succesfull..')">Reset</button>
        <button type="submit" onclick="alert('Submitted
Succesfully..')">submit</button>
    </td>
</tr>
</table>
</form>
</body>
</html>

```

OUTPUT:

COLLEGE ADMINISTRATION REGISTRATION	
FULL NAME	Fullname
E-mail	abcd@gmail.com
Phone Number	Phone Number
Gender	☐ Male ☐ Female ☐ Other
Course Applying for	--Select--
Hobbies	☐ Sports ☐ Music ☐ Reading
Date	dd-mm-yyyy 
Upload Certificates	Choose Files No file chosen
Address	Enter your address here!!
Reset submit	

Explanation:

1. `<!DOCTYPE html>` defines the document type and tells the browser this is an HTML5 page.
2. The `<html lang="en">` tag starts the HTML document and sets the language to English.
3. Inside `<head>`, the `<meta charset="UTF-8">` ensures proper character encoding.
4. `<meta name="viewport">` makes the page responsive on mobile devices.
5. The `<title>` tag sets the page title as student form (shown in browser tab).

6. Inside `<style>`, CSS is used to set the body background color and table styling.
7. The `<form>` tag creates a form to collect user input for college registration.
8. A `<table>` is used to align labels and input fields in rows and columns.
9. Various input types are used: text, email, number, radio, checkbox, date, and file.
10. The `<select>` dropdown allows students to choose a course (B.Tech, M.Tech, M.B.A, Other).
11. The `<textarea>` tag is used to enter a multi-line address.
12. The Reset button clears the form and shows an alert message, while the Submit button shows a success alert when clicked.

4B. A Company wants to design a webpage that displays two different forms side by side using iframes. one iframe should display a login form and the other should display a registration form.

```
<!-- AV.SC.U4CSE25136 RITHVIK -->
<!DOCTYPE html>
<html>
<head>
  <title>Document</title>
  <style>
    td{
      padding: 6px;
      font-size: large;
    }
    div{
      width: 400px;
      font-size: larger;
      background-color: rgb(193, 255, 244);
    }
    button{
      background-color: rgb(41, 115, 4);
      color: antiquewhite;
      font-size: larger;
    }
    body{
      padding-left: 180px;
      padding-top: 60px;
    }
  </style>
</head>
<body>
  <form>
    <div>

      <table border="2" align="center" style="border-collapse: collapse;"
width="300" height="170">
        <h3 style="text-align: center;">USER LOGIN</h3>
        <tr>
          <td> <label for="useranme">Username:</label> </td>
          <td> <input type="text" id="useranme" placeholder="username"
required> </td>
        </tr>

        <tr>
          <td> <label for="password">Password</label></td>
          <td> <input type="password" id="password"
placeholder="password"required></td>
```

```

        </tr>

        <tr style="height: 30">
            <td colspan="2" style="padding-left: 100px;"> <button
type="button" onclick="alert('Login succesfull')" ;>Login</button></td>
        </tr>

    </table>
</div>
</form>
</body>
</html>

```

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
<style>
    td{
        padding: 6px;
        font-size: large;
    }

    div{
        width: 500px;
        font-size: larger;
        background-color: rgb(193, 255, 244);
    }
    .reset{
        background-color: rgb(220, 20, 20);
        color: antiquewhite;
    }
    .submit{
        background-color: rgb(41, 115, 4);
        color: antiquewhite
    }
    button{
        font-size: large;
    }

```

```

    body{
        padding-left: 120px;
        padding-top: 60px;
    }
</style>

</head>
<body>
    <form>
    <div>
        <table align="center" border="2" style="border-collapse:
collapse;padding:10px" height="400" width="400">

            <h3 style="text-align: center;">USER REGISTRATION</h3>

            <tr>
                <td> <label for="fname">FULL NAME:</label> </td>
                <td> <input type="text" id="fname" placeholder="Fullname">
</td>
            </tr>

            <tr>
                <td> <label for="email">E-mail</label> </td>
                <td> <input type="email" id="email"
placeholder="abcd@gmail.com"> </td>
            </tr>

            <tr>
                <td> <label for="phone">Phone</label> </td>
                <td> <input type="number"
id="phone" placeholder="Ex:1234567890"> </td>
            </tr>

            <tr>
                <td> <label for="password">Password</label> </td>
                <td> <input type="password" id="password"
placeholder="Password">
            </tr>

            <tr>
                <td colspan="2" style="padding-left: 110px;">
                    <button type="reset" class="reset" onclick="alert('reset
succesfully')">Reset</button>
                    <button type="submit" class="submit"
onclick="alert('Submitted succesfully')">Submit</button>
                </td>
            </tr>
        </table>

```

```
    </div>
  </form>
</body>
</html>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>

  <style>
    body{
      background-color: blanchedalmond;
    }
    .box{
      display: flex;
      height: 100vh;
    }
    iframe{
      flex: 1;
    }
  </style>
<body>
  <div class="box">
    <iframe
      src="login.html"
      title="login details" >

    </iframe>

    <iframe
      src="register.html"
      title="registration details"

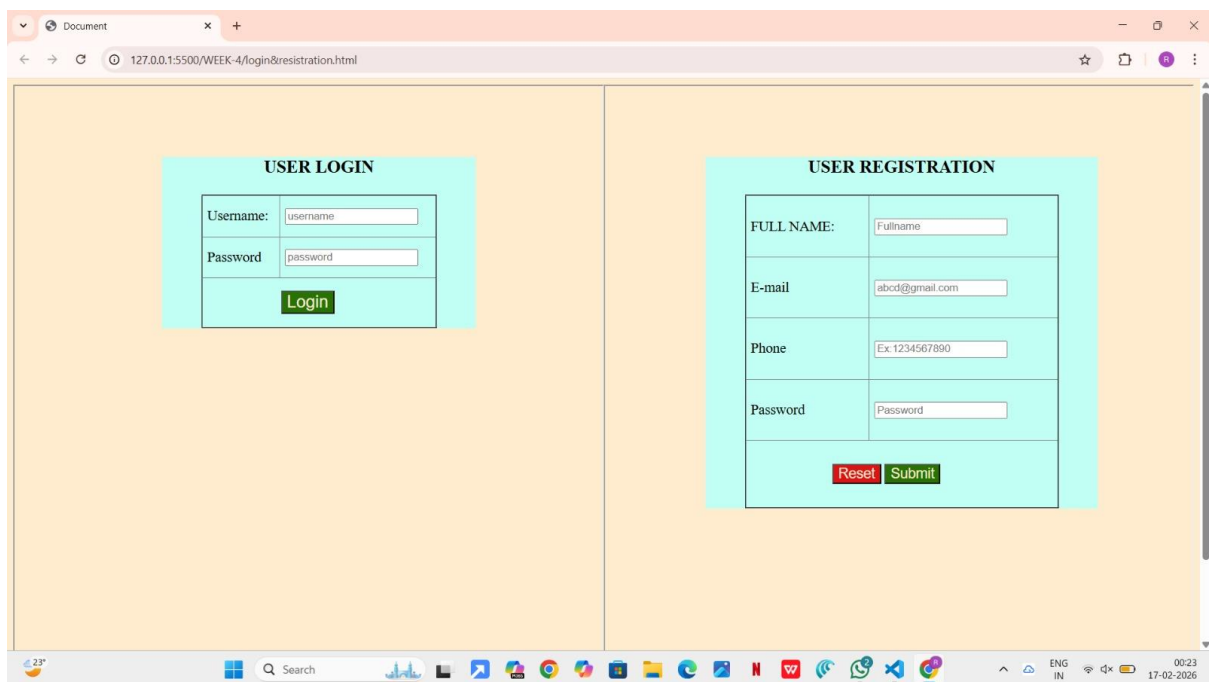
    >
    </iframe>
  </div>
```

```

<marquee direction="right" scrollamount="10" width="100%" height="100px">
    Note: Every student register and Login into this application
</marquee>
</body>
</html>

```

OUTPUT:



Explanation:

1..Login Page

1. The page starts with <!DOCTYPE html> which defines the document as HTML5.
2. <form> tag is used to collect user input.
3. A <div> container is used to give background color and fixed width.

4. A <table> is used to align the Username and Password fields in rows and columns.

5. CSS is applied:

td → Adds padding and increases font size.

div → Sets width (400px) and background color.

button → Styles button with green background and light text.

body → Adds left and top spacing.

6. <input type="text" required> → Takes username input (mandatory field)

7. <input type="password" required> → Takes password input (hidden text).

8. <button type="button"> → Does not submit form.

9. onclick="alert('Login successful')" → Displays popup message.

10. No backend or real authentication is implemented.

2..Registration Page

1. Another HTML document with a registration form.

2. Uses <form>, <div>, and <table> for structure.

3. CSS styling includes:

.reset class → Red reset button.

.submit class → Green submit button.

button → Increases font size.

body → Adds spacing.

4. Input fields:

Full Name → type="text"

Email → type="email" (validates email format)

Phone → type="number"

Password → type="password"

5. <button type="reset"> → Clears all form fields.
6. <button type="submit"> → Submits the form.
7. Both buttons show alert messages using onclick.
8. No backend connection or data storage.

3.. Main Page (Iframe Page)

1. Uses <!DOCTYPE html> and standard HTML structure.
2. CSS:
display: flex; → Places elements side-by-side.
height: 100vh; → Full screen height.
iframe { flex: 1; } → Both iframes get equal width.
3. First <iframe> loads login.html.
4. Second <iframe> loads register.html.
5. <marquee> tag is used to scroll text horizontally.
6. <marquee> is outdated and not recommended in modern HTML.

WEEK-5

5(A) Aim: To create a responsive card layout using HTML and CSS with three cards, each containing an image, title, and button.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-  
scale=1.0">
```

```
  <title>Document</title>
```

```
<style>
```

```
  body{
```

```
    min-height: 100vh;
```

```
  }
```

```
  .heros{
```

```
    display: flex;
```

```
    gap:50px;
```

```
    margin: 20px;
```

```
  }
```

```
  .heros img {
```

```
    width:200;
```

```
    height: 250px;
```

```
    display: block;
```

```
    margin: auto;
```

```
}
```

```
.heros button{  
    background-color: rgb(109, 106, 97,0);  
    color: rgb(36, 4, 217);  
    display: block;  
    margin: auto;
```

```
}
```

```
.aa{  
    margin: 10px;  
    padding: 10px;  
}
```

```
.aa:hover{  
    transform: scale(1.1);  
    background-color: rgba(178, 178, 178, 0.308);  
    backdrop-filter: blur(5px);  
    border-radius: 15px;  
    margin: 10px;  
    padding: 10px;  
    transition: transform 0.5s ease;
```

```
}  
.bb{  
    position:fixed;  
    top: 0;  
    left: 0;  
    z-index:-1;  
    width:100%;  
    height:100%;  
    object-fit:cover;  
}  
p{  
    color: aliceblue;  
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;  
}  
h1{  
    font-family: 'Courier New', Courier, monospace;  
    color: rgb(255, 0, 0);  
    font-weight: bolder;  
    font-size: xx-large;  
}  
audio{  
    background-color: rgb(161, 51, 51,0);  
    opacity:0;  
}  
h3{
```

```

        color: rgb(255, 0, 0);
        text-align: center;
        font-family: 'Courier New', Courier, monospace;
        font-weight: bolder;
        font-size: x-large;
    }
</style>
</head>
<body>  <audio autoplay controls >

    <source src="Rakta Charitra Dorikite Chastavu-Remix - DJ ROCK
BEATS.mp3" type="audio/mp3">

    <h1 style="text-align: center;">tollywood heros</h1>
</audio>

<video class="bb" autoplay loop controls muted playsinline>

    <source src="Fire_72___45s___4k_res.mp4" type="video/mp4">
</video>
<div class="heros">

    <div class="aa">

        <h3>REBEL ★ STAR</h3>

        <p>Uppalapati Venkata Suryanarayana Prabhas Raju ([prabʰa:s];
born 23 October 1979), known mononymously as Prabhas, is an Indian
actor who predominantly works in Telugu cinema.[4] He is one of the
highest-paid actors in Indian cinema and has been featured in Forbes
India's Celebrity 100 list since 2015.[9] Referred to in the media as Rebel

```

Star, he has appeared in over 24 films, and has received seven Filmfare Awards nominations, a Nandi Award, and a SIIMA Award.[10][11]</p>

<button>Learn more</button>

</div>

<div class="aa">

<h3>POWER★STAR</h3>

<p>Konidela Pawan Kalyan[5] (born Konidela Kalyan Babu;[8] 2 September 1971)[2] is an Indian politician, actor, philanthropist, and martial artist serving as the 11th Deputy Chief Minister of Andhra Pradesh since June 2024. He is also the Minister of Panchayat Raj, Rural Development and Rural Water Supply; Environment, Forest, Science and Technology in the Government of Andhra Pradesh as MLA representing the Pithapuram constituency.[3] He is the founding president of the Janasena Party</p>

<button>Learn more</button>

</div>

<div class="aa">

<h3>YOUNG🐅TIGER</h3>

<p>Nandamuri Taraka Rama Rao (born 20 May 1983), popularly known as NTR Jr, is an Indian actor, producer, and television presenter who primarily works in Telugu cinema.[1] He is one of the highest-paid actors in Indian cinema and has been featured in Forbes India's Celebrity 100 list since 2012.[2][3] Referred to in the media as the Man of Masses, he has

starred in over 30 films.[4] He is a recipient of several accolades including, three Filmfare Awards South, three CineMAA Awards, a SIIMA Award, a IIFA Award and two Nandi Awards.</p>

<button>Learn more</button>

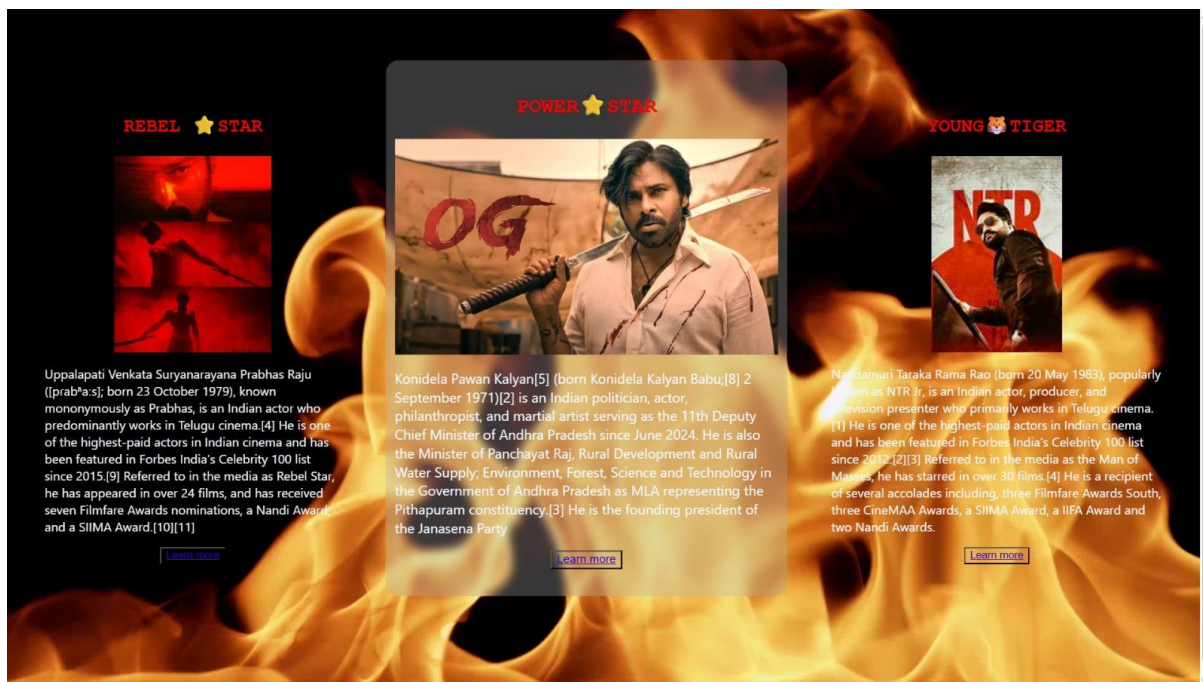
</div>

</div>

</body>

</html>

OUTPUT:



EXPLANATION:

1. The `<!DOCTYPE html>` declaration tells the browser that the document is an HTML5 webpage.
2. The `<html lang="en">` tag starts the HTML document and specifies that the language is English.
3. Inside `<head>`, the `<meta>` tags define character encoding (UTF-8) and make the page responsive on different devices.
4. The `<style>` section contains CSS used to design the page layout, colors, fonts, hover effects, and positioning.
5. The `body` is given a minimum height of the full screen using `min-height: 100vh`.
6. The `.heros` class uses Flexbox (`display: flex`) to arrange the hero cards horizontally with space (`gap: 50px`).
7. The `.aa` class represents each hero card, and the `:hover` effect enlarges the card with `transform: scale(1.1)` and adds blur and background styling.
8. The `.bb` class styles the background video using `position: fixed` and `object-fit: cover` so it fills the entire screen behind the content.
9. The `<audio>` tag automatically plays background music when the page loads.
10. The `<video>` tag plays a looping muted video as the page background.
11. Each hero section contains a heading (`h3`), image (`img`), and description (`p`) about the actor.
12. The “Learn more” button is inside an `<a>` tag that opens the actor’s Wikipedia page in a new tab using `target="_blank"`.

5(B) Aim: To design a mini website using iframe with navigation links (Home, About, Contact), where clicking on each link displays the respective content within the same page using internal or external CSS.

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-
scale=1.0">

    <title>Document</title>

</head>

<style>

    span h2{

        font-family: 'Courier New', Courier, monospace;

        border: 3px solid;

        border-radius: 10px;

        padding: 10px;

        background-color: chartreuse;

        color: red;

        text-align: center;

        font-size: xx-large;

    }

    button {

        text-decoration: solid;
```

```

padding: 16px;
border-radius:15px;
font-size: larger;
background-color:rgb(0, 229, 255) ;
border-color: hotpink;
}
nav{
text-align: center;
border: 3px;
padding:9px;
background-color: rgb(188, 172, 180);
}
nav img{
width: 400;
height: 60;

}
</style>
<body>
<span><h2>SPORTS BIKES </h2></span>
<nav>
<a href="BIKE1.html" target="home"><button>B M W </button></a>
<a href="bike2.html" target="home"><button>ZX 10R</button></a>
<a href="bike3.html" target="home"><button>ENFIELD</button></a>
</nav>

```

```
<iframe src="BIKE1.html" name="home" title="home" height="600px"
width="100%"></iframe>
```

```
</html>
```

BIKE1.HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
```

```
    <title>Document</title>
```

```
</head>
```

```
<style>
```

```
    img{
```

```
        display: block;
```

```
        margin: auto;
```

```
    }
```

```
</style>
```

```
<body>
```

```
    
```

```
</body>
```

```
</html>
```

Bike2.html

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Document</title>
</head>
<style>
  img{
    display: block;
    margin: auto;
  }
</style>
<body>
  
</body>
</html>
```

Bike3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
```

```

<title>Document</title>

</head>

<style>

  img{

    display: block;

    margin: auto;

  }

</style>

<body>

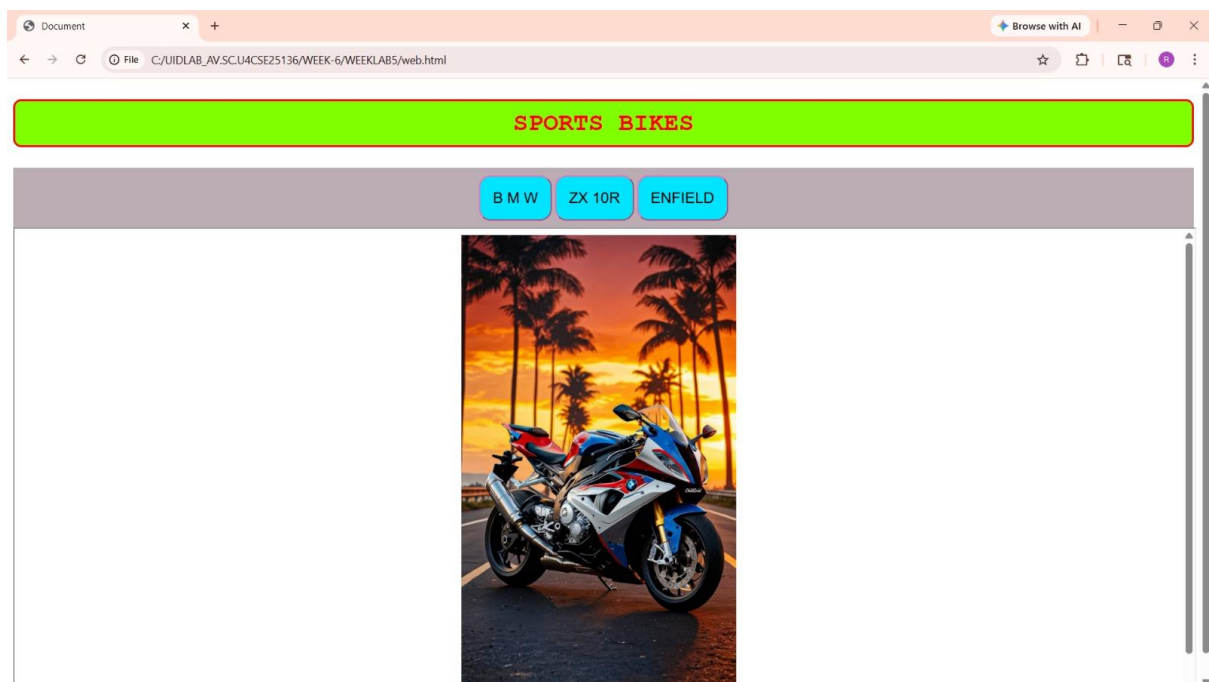
  

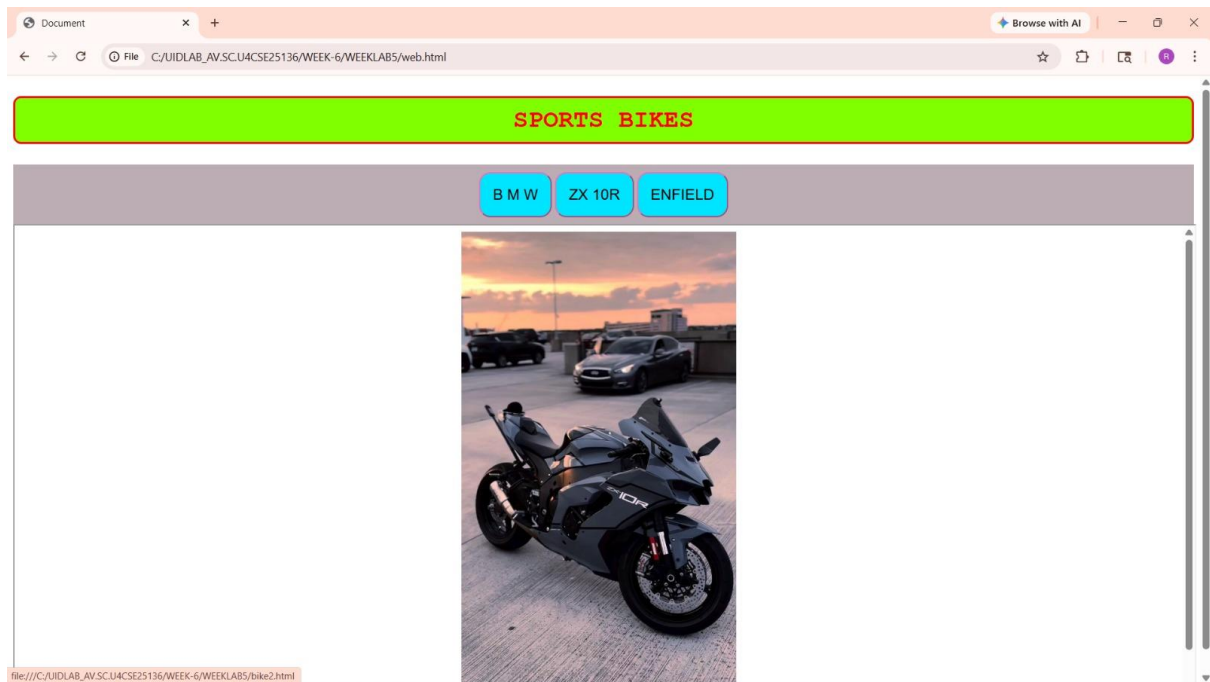
</body>

</html>

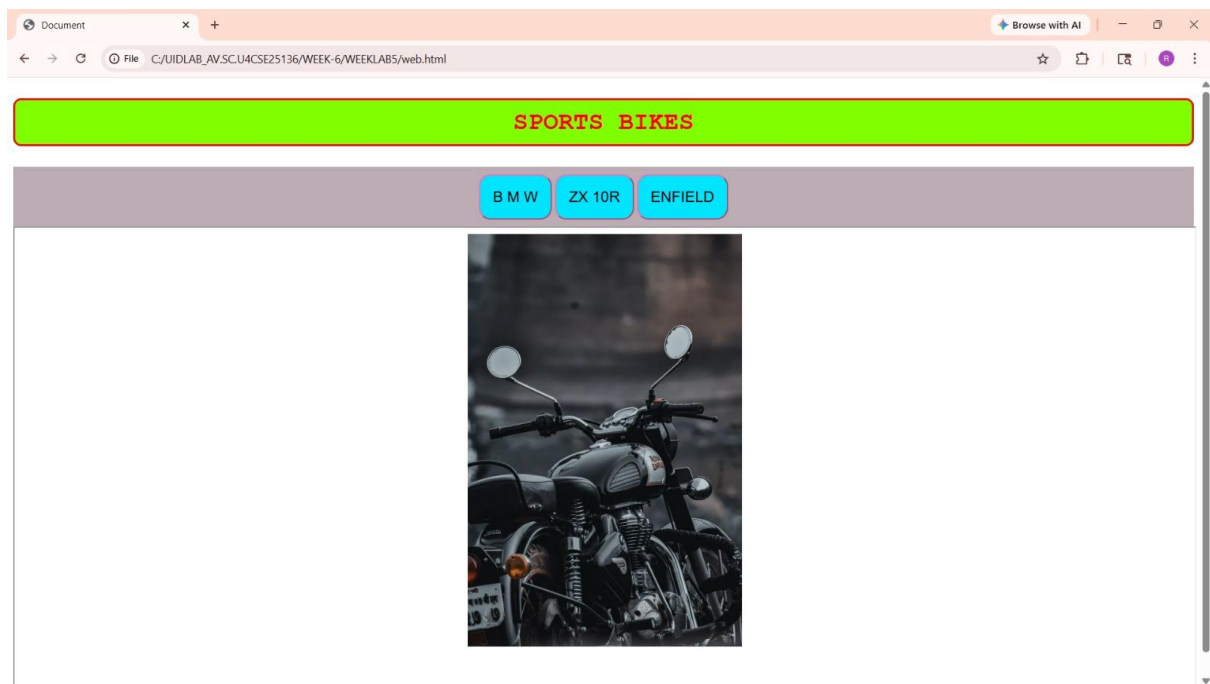
```

OUTPUT:





X



EXPLANATION:

1. `<!DOCTYPE html>` declares the document as HTML5.
2. `<html lang="en">` is the root element and sets the page language to English.
3. The `<head>` section contains metadata like character set, viewport, and title.
4. `<meta charset="UTF-8">` allows the browser to display **all** characters properly.
5. `<style>` contains CSS used to design the webpage.
6. `span h2` styles the heading with **border**, background color, large font, and centered text.
7. `button` styling adds padding, rounded corners, font size, and background color.
8. `<nav>` creates a navigation bar with buttons linking to different bike pages.
9. Each `<a>` link uses `target="home"` to load the page inside the iframe.
10. `<iframe>` displays another webpage inside the current page, showing `BIKE1.html` by default.

5(C) Aim: To design an Amrita login page using HTML and CSS that displays different campuses, where clicking on each campus shows its details on the same page.

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-
scale=1.0">

    <title>Document</title>

    <style>

        .aa{

            display: flex;

            height:100vh;

        }

        h2{

            border: 3px solid;

            text-align: center;

            font-family: 'Courier New', Courier, monospace;

            padding: 15px;

            border-radius: 50px;

            background-color: cadetblue;

            color: antiquewhite;

        }
```

```
nav{
    border:2px solid;
    text-align: center;
    font-family: 'Times New Roman', Times, serif;
    padding: 10px;
    border-radius: 12px;
    background-color: hotpink;
    color: forestgreen;

}
nav a{
    text-decoration: none;
    color: black;
}
</style>
</head>
<body>
<h2>AMRITA VISHWA VIDHYAPEETHAM</h2>
<nav>
    <a href="site.html" target="site">AMARAVATI</a>
    <a href="bengaluru.html" target="site">BENGALURU</a>
    <a href="coimbatore.html" target="site">COIMBATORE</a>
    <a href="chennai.html" target="site">CHENNAI</a>
    <a href="amritapuri.html" target="site">AMRITAPURI</a>
</nav>
```

```
<div class="aa">
```

```
<iframe src="login.html" title="login"></iframe>
```

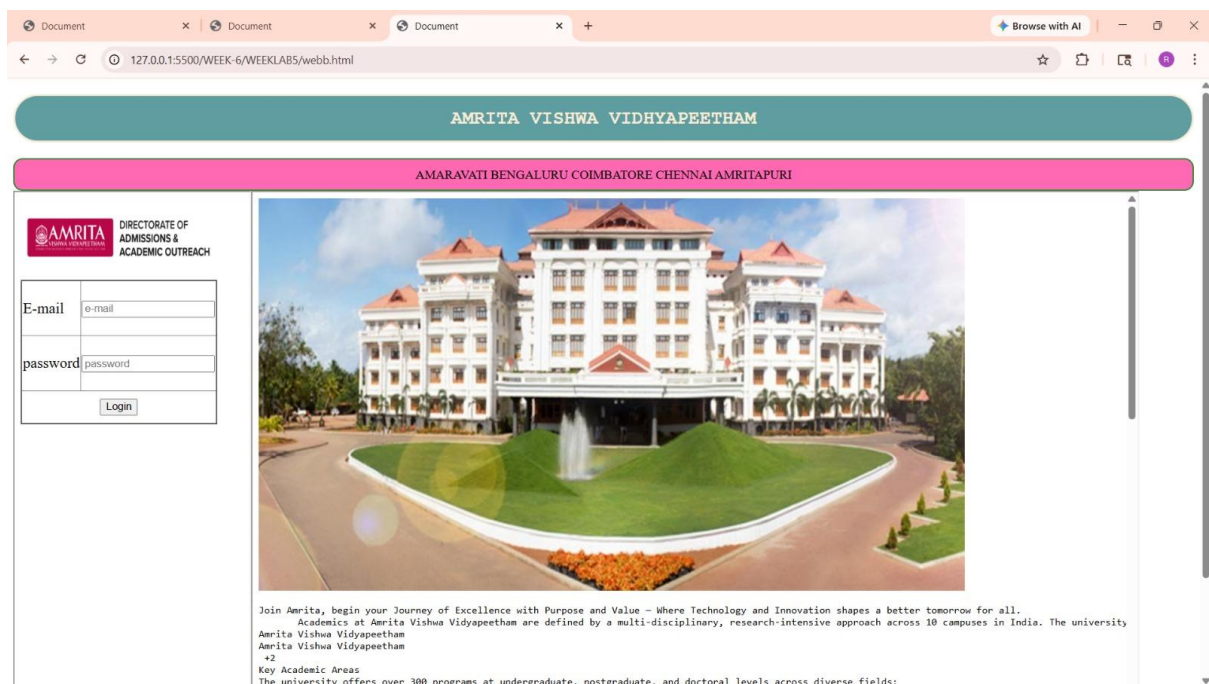
```
<iframe src="site.html" title="site" name="site" width="75%"></iframe>
```

```
</div>
```

```
</body>
```

```
</html>
```

OUTPUT:



Document
Document
Document
+
Browse with AI
127.0.0.1:5500/WEEK-6/WEEKLAB5/webb.html

AMRITA VISHWA VIDHYAPEETHAM

AMARAVATI BENGALURU COIMBATORE CHENNAI AMRITAPURI


DIRECTORATE OF ADMISSIONS & ACADEMIC OUTREACH

E-mail:
password:



Join Amrita, begin your Journey of Excellence with Purpose and Value – Where Technology and Innovation shapes a better tomorrow for all.

Academics at Amrita Vishwa Vidyapeetham are defined by a multi-disciplinary, research-intensive approach across 10 campuses in India. The university, Amrita Vishwa Vidyapeetham

Amrita Vishwa Vidyapeetham

+2

Key Academic Areas

The university offers over 300 programs at undergraduate, postgraduate, and doctoral levels across diverse fields:

Document
Document
Document
+
Browse with AI
127.0.0.1:5500/WEEK-6/WEEKLAB5/webb.html

AMRITA VISHWA VIDHYAPEETHAM

AMARAVATI BENGALURU COIMBATORE CHENNAI AMRITAPURI


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Amrita Vishwa Vidyapeetham

+2

Key Academic Areas

The university offers over 300 programs at undergraduate, postgraduate, and doctoral levels across diverse fields:

Document
Document
Document
+
Browse with AI
127.0.0.1:5500/WEEK-6/WEEKLAB5/webb.html

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Amrita Vishwa Vidyapeetham

+2

Key Academic Areas

The university offers over 300 programs at undergraduate, postgraduate, and doctoral levels across diverse fields:

Document
Document
Document
+
Browse with AI
127.0.0.1:5500/WEEK-6/WEEKLAB5/webb.html

AMRITA VISHWA VIDHYAPEETHAM

AMARAVATI BENGALURU COIMBATORE CHENNAI AMRITAPURI


DIRECTORATE OF ADMISSIONS & ACADEMIC OUTREACH

E-mail:
password:



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Amrita Vishwa Vidyapeetham

+2

Key Academic Areas

The university offers over 300 programs at undergraduate, postgraduate, and doctoral levels across diverse fields:

EXPLANATION:

1. `<!DOCTYPE html>` tells the browser that the document is written in HTML5.
2. `<html lang="en">` starts the HTML document and specifies that the language of the webpage is English.
3. Inside the `<head>` section, meta tags are used to define character encoding (`UTF-8`) and make the page responsive using the viewport tag. The `<title>` sets the page title.
4. The CSS inside `<style>` defines the design of the webpage. The class `.aa` uses `display: flex` so its elements appear side by side and it takes the full screen height (`100vh`).
5. The `h2` style adds a border, center alignment, padding, rounded corners, background color (cadetblue), and text color to the heading.
6. The `nav` style is used to design the navigation bar with border, padding, rounded corners, background color (hotpink), and font style.
7. `nav a` removes the default underline from links and sets the link color to black.
8. In the body, `<h2>` displays the heading **“AMRITA VISHWA VIDHYAPEETHAM”** at the top of the webpage.
9. The `<nav>` section contains hyperlinks for different campus pages like Amaravati, Bengaluru, Coimbatore, Chennai, and Amritapuri. When clicked, they open inside the iframe named **site**.
10. The `<div class="aa">` contains two iframes: the first loads `login.html`, and the second loads `site.html`. Because of flex layout, both frames appear side by side on the page.

WEEK-6

Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

MAIN CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="style.css">
</head>
<style>
video{
  position:fixed;
  top: 0;
  left: 0;
  z-index: -1;
  width: 100%;
```

```
height: 100%;  
object-fit: fill;  
}
```

```
</style>
```

```
<body>
```

```
<div class="container">
```

```
<video controls muted autoplay loop>
```

```
<source src="videofashion.mp4" >
```

```
</video>
```

```
<div class="head"> <h2>Flipkart</h2> </div>
```

```
<div class="sale"><h4>Sale starts from march 6 to 7.. Upto 80%  
offer🔥🔥🔥<h4></div>
```

```
<div class="menu">
```

```
<a href="allcategories.html" target="visible" class="link"><h3>All  
categories</h3><hr></a>
```

```
<a href="fashion.html" target="visible"  
class="link"><h3>Fashion👕</h3><hr></a>
```

```
<a href="moblies.html" target="visible"  
class="link"><h3>Mobiles📱</h3><hr></a>
```

```
<a href="beauty.html" target="visible"  
class="link"><h3>Beauty💄</h3><hr></a>
```

```
<a href="electronics.html" target="visible"  
class="link"><h3>Electronics💻</h3><hr></a>
```

```
</div>
```



```
<div class="allcategories">  
    <iframe src="allcategories.html" title="demo" name="visible"  
></iframe>  
</div>
```

```
<div class="foot">  
    <div><h3>ABOUT</h3>  
        Contact Us  
        About Us  
        Careers  
        Flipkart Stories  
        Press  
        Corporate Information  
    </div>
```

```
<div><h3>HELP</h3>  
    Payments  
    Shipping  
    Cancellation & Returns  
    FAQ  
</div>
```

```
<div><h3>CONSUMER POLICY</h3>
```

Cancellation & Returns

Terms Of Use

Security

Privacy

Sitemap

Grievance Redressal

EPR Compliance

FSSAI Food Safety Connect App

</div>

<div><h3>Registered Office Address</h3>

Flipkart Internet Private Limited,

Buildings Alyssa, Begonia &

Clove Embassy Tech Village,

Outer Ring Road, Devarabeesanahalli Village,

Bengaluru, 560103,

Karnataka, India

CIN : U51109KA2012PTC066107

Telephone: 044-45614700 / 044-67415800

</div>

</div>

</div>

</body>

</html>

STYLE.CSS FILE:

```
.container{  
  
    display: grid;  
    grid-template-areas:  
        "head head"  
        "alert alert"  
        "nav site"  
        "foot foot";  
    grid-template-columns: 15% 1fr;  
    grid-template-rows:auto auto 70% auto;  
}
```

```
.head{  
    grid-area: head;  
    border: 3px solid ;  
    font-weight: bold;  
    font-family: 'Courier New', Courier, monospace;  
    text-align: center;  
    margin: 4px;  
    background-color: rgba(95,95,95, 0.5);
```

```

    color: rgb(0, 252, 185);
    border-radius: 15px;
    font-size: larger;
    backdrop-filter: blur(10px);
}

.sale{
    grid-area:alert;
    border: 3px solid ;
    font-weight: bold;
    font-family: 'Courier New', Courier, monospace;
    text-align: center;
    margin: 4px;
    color: rgb(0, 252, 185);
    background-color: rgba(95,95,95,0.5);
    backdrop-filter: blur(10px);
    border-radius: 15px;
    font-size: larger;
}

.menu{
    grid-area:nav;
    border: 2px solid rgb(0, 252, 185);;
    margin:10px 25px 0px 15px;
    padding: 4px;
    background-color:rgb(95, 95, 95,0.5);

```

```
border-radius: 15px;

backdrop-filter: blur(10px);
}

.menu h3{
color: rgb(0, 252, 185);
}

iframe{

grid-area: site;

width:100%;

height: 96%;

margin:10px 0px 0px 0px;

border-radius: 15px;
}

.foot{

grid-area: foot;

display:flex;

gap:25px;

border: 2px solid ;

font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida
Sans Unicode', Geneva, Verdana, sans-serif;

text-align: center;

margin: 4px;

align-items: center;

color: rgb(230, 255, 255);
```

```
background-color: rgb(95,95,95,0.4);
backdrop-filter: blur(10px);
font-size: medium;
padding: 15px;
border-radius: 20px;

}

.foot h3{
    color: rgb(0, 252, 185);
}

.link{
    text-decoration:none;
    color:black
}
```

ALLCATEGORIES.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Document</title>
```

```
</head>

<body>

  <style>

    body{

      text-align: center

      ;

    }

    .container{

      display:flex;

    }

    .block1{

      padding: 10px;

      margin: 10px;

      border:2px solid;

      border-radius: 15px;

    }

    .block2{

      padding: 10px;

      margin: 10px;

      border:2px solid;

      border-radius: 15px;

    }

    .block3{

      padding: 10px;

      margin: 10px;
```

```
border:2px solid;
border-radius: 15px;
}
.block4{
padding: 3px;
margin: 10px;
border:2px solid;
border-radius: 15px;
}
.container img{
display: block;
margin:auto;
align-items: center;
justify-content: center;
width:250px;
height:300px;
}
.block1:hover,.block2:hover,.block3:hover,.block4:hover{
transform: scale(1.1);
transition: transform 0.5s ease;
}
video{
position: fixed;
top:0;
left: 0;
```



```

        z-index: -1;

        width: 100%;

        height: 100%;

        object-fit: fill;
    }

    .block1 ,.block2 ,.block3 ,.block4{

        background-color: rgba(255, 255, 255, 0.3);

        backdrop-filter: blur(15px);
    }
</style>
<body>

    <video controls autoplay muted loop>

        <source src="videofashion.mp4">

    </video>

    <div class="container">

        <div class="block1">

            <a href="fashion.html"></a>

        </div>

        <div class="block2">

            <a href="moblies.html"></a>

        </div>

        <div class="block4">

```

```
<a href="beauty.html">  </a>

</div>

<div class="block3">

    <a href="electronics.html">  </a>

</div>

</div>

</body>

</html>
```

FASHION.HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-
scale=1.0">

    <title>Document</title>

</head>

<body>

    <style>

        body {
```

```
    text-align: center;
}

.container{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(300px,1fr));
    gap: 40px;
}

.block1{
    padding: 10px;
    margin: 10px;
    border:2px solid;
    border-radius: 15px;
}

.block2{
    padding: 10px;
    margin: 10px;
    border:2px solid;
    border-radius: 15px;
}

.block3{
    padding: 10px;
    margin: 10px;
    border:2px solid;
    border-radius: 15px;
}
```

```
.block4{
    padding: 3px;
    margin: 10px;
    border:2px solid;
    border-radius: 15px;
}

.container img{
    margin:auto;
    align-items: center;
    justify-content: center;
    width:270px;
    height:300px;
}

.block1:hover,.block2:hover,.block3:hover,.block4:hover{
    transform: scale(1.1);
    transition: transform 0.5s ease;
}

s{
    color: red;
}

pre{
    text-align: center;
    font-size: larger;
    font-weight: bold;
}
```

```

video{
    position:fixed;
    top: 0;
    left: 0;
    z-index: -1;
    width: 100%;
    height: 100%;
    object-fit: fill;
}

.block1 ,.block2 ,.block3 ,.block4{
    background-color: rgba(255, 255, 255, 0.3);
    backdrop-filter: blur(15px);
}

</style>

<body>

    <div class="container">

        <video controls muted autoplay loop>

            <source src="videofashion.mp4" >

        </video>

        <div class="block1">

            <pre> $499 <s>$874</s></pre>

        </div>

        <div class="block2">

            <pre> $599 <s>$1074</s></pre>

```

```
</div>

<div class="block3">

    <pre> $899 <s>$1389</s></pre>

  </div>

<div class="block4">

    <pre> $1099 <s>$1699</s></pre>

  </div>

</div>

</body>

</html>
```

MOBILES.HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-
scale=1.0">

  <title>Document</title>
```

```
</head>

<body>

  <style>

    body{

      text-align: center;

    }

    .container{

      display:grid;

      grid-template-columns:repeat(auto-fit,minmax(300px,1fr));

      gap: 40px;

    }

    .block1{

      padding: 10px;

      margin: 10px;

      border:2px solid;

      border-radius: 15px;

    }

    .block2{

      padding: 10px;

      margin: 10px;

      border:2px solid;

      border-radius: 15px;

    }

    .block3{

      padding: 10px;
```

```

    margin: 10px;
    border: 2px solid;
    border-radius: 15px;
}
.block4{
    padding: 3px;
    margin: 10px;
    border: 2px solid;
    border-radius: 15px;
}
.container img{
    align-items: center;
    justify-content: center;
    width: 250px;
    height: 300px;
}
.block1:hover,.block2:hover,.block3:hover,.block4:hover{
    transform: scale(1.1);
    transition: transform 0.5s ease;
}
s{
    color: red;
}
pre{
    text-align: center;

```



```

        font-size: larger;
        font-weight: bold;
    }
    video{
        position:fixed;
        top: 0;
        left: 0;
        z-index: -1;
        width: 100%;
        height: 100%;
        object-fit: fill;
    }
    .block1 ,.block2 ,.block3 ,.block4{
        background-color: rgba(255, 255, 255, 0.3);
        backdrop-filter: blur(15px);
    }
</style>
<body>
    <video controls muted autoplay loop>
    <source src="videofashion.mp4" >
    </video>
    <div class="container">
        <div class="block1">
            <pre> $4499 <s>$7874</s></pre>

```

</div>

<div class="block2">

<pre
> \$19599 <s>\$21074</s></pre>

</div>

<div class="block3">

<pre> \$1899 <s>\$1989</s></pre>

</div>

<div class="block4">

<pre> \$11099 <s>\$13699</s></pre>

</div>

</div>

</body>

</html>

BEAUTY.HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-
scale=1.0">

  <title>Document</title>

</head>

<body>

  <style>

    body{

      text-align: center;

    }

    .container{

      display:grid;

      grid-template-columns:repeat(auto-fit,minmax(300px,1fr));

      gap: 40px;

    }

    .block1{

      padding: 10px;

      margin: 10px;

      border:2px solid;

      border-radius: 15px;
```

```
}  
.block2{  
    padding: 10px;  
    margin: 10px;  
    border:2px solid;  
    border-radius: 15px;  
}  
.block3{  
    padding: 10px;  
    margin: 10px;  
    border:2px solid;  
    border-radius: 15px;  
}  
.block4{  
    padding: 3px;  
    margin: 10px;  
    border:2px solid;  
    border-radius: 15px;  
}  
.container img{  
    align-items: center;  
    justify-content: center;  
    width:250px;  
    height:300px;  
}
```

```
.block1:hover,.block2:hover,.block3:hover,.block4:hover{
    transform: scale(1.1);
    transition: transform 0.5s ease;
}
s{
    color: red;
}
pre{
    text-align: center;
    font-size: larger;
    font-weight: bold;
}
video{
    position:fixed;
    top: 0;
    left: 0;
    z-index: -1;
    width: 100%;
    height: 100%;
    object-fit: fill;
}
.block1 ,.block2 ,.block3 ,.block4{
    background-color: rgba(255, 255, 255, 0.3);
    backdrop-filter: blur(15px);
}
```

```
</style>
<body>
  <video controls muted autoplay loop>
    <source src="videofashion.mp4" >
  </video>
  <div class="container">
    <div class="block1">
      <pre> $99 <s>$174</s></pre>
    </div>
    <div class="block2">
      <pre>
$199 <s>$210</s></pre>
    </div>
    <div class="block3">
      
      <pre> $189 <s>$198</s></pre>
    </div>
    <div class="block4">
      
      <pre> $110 <s>$136</s></pre>
    </div>
  </div>
```

```
</body>
```

```
</html>
```

ELECTRONICS.HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-  
scale=1.0">
```

```
  <title>Document</title>
```

```
</head>
```

```
<body>
```

```
  <style>
```

```
    body{
```

```
      text-align: center;
```

```
    }
```

```
    .container{
```

```
      display:grid;
```

```
      grid-template-columns:repeat(auto-fit,minmax(300px,1fr));
```

```
      gap: 40px;
```

```
    }
```

```
    .block1{
```

```
      padding: 10px;
```

```
margin: 10px;
border: 2px solid;
border-radius: 15px;
}
.block2{
padding: 10px;
margin: 10px;
border: 2px solid;
border-radius: 15px;
}
.block3{
padding: 10px;
margin: 10px;
border: 2px solid;
border-radius: 15px;
}
.block4{
padding: 3px;
margin: 10px;
border: 2px solid;
border-radius: 15px;
}
.container img{
align-items: center;
justify-content: center;
```



```
width:250px;
height:300px;
}
.block1:hover,.block2:hover,.block3:hover,.block4:hover{
    transform: scale(1.1);
    transition: transform 0.5s ease;
}
s{
    color: red;
}
pre{
    text-align: center;
    font-size: larger;
    font-weight: bold;
}
video{
    position:fixed;
    top: 0;
    left: 0;
    z-index: -1;
    width: 100%;
    height: 100%;
    object-fit: fill;
}
.block1 ,.block2 ,.block3 ,.block4{
```

```

        background-color: rgba(255, 255, 255, 0.3);
        backdrop-filter: blur(15px);
    }
</style>
<body>
    <video controls muted autoplay loop>
    <source src="videofashion.mp4" >
    </video>
    <div class="container">
        <div class="block1">
            <pre> $5299 <s>$8374</s></pre>
        </div>
        <div class="block2">
            <pre> $199 <s>$210</s></pre>
        </div>
        <div class="block3">
            
            <pre> $589 <s>$698</s></pre>
        </div>
        <div class="block4">
            

```

```

    <pre> $910 <s>$1136</s></pre>

  </div>

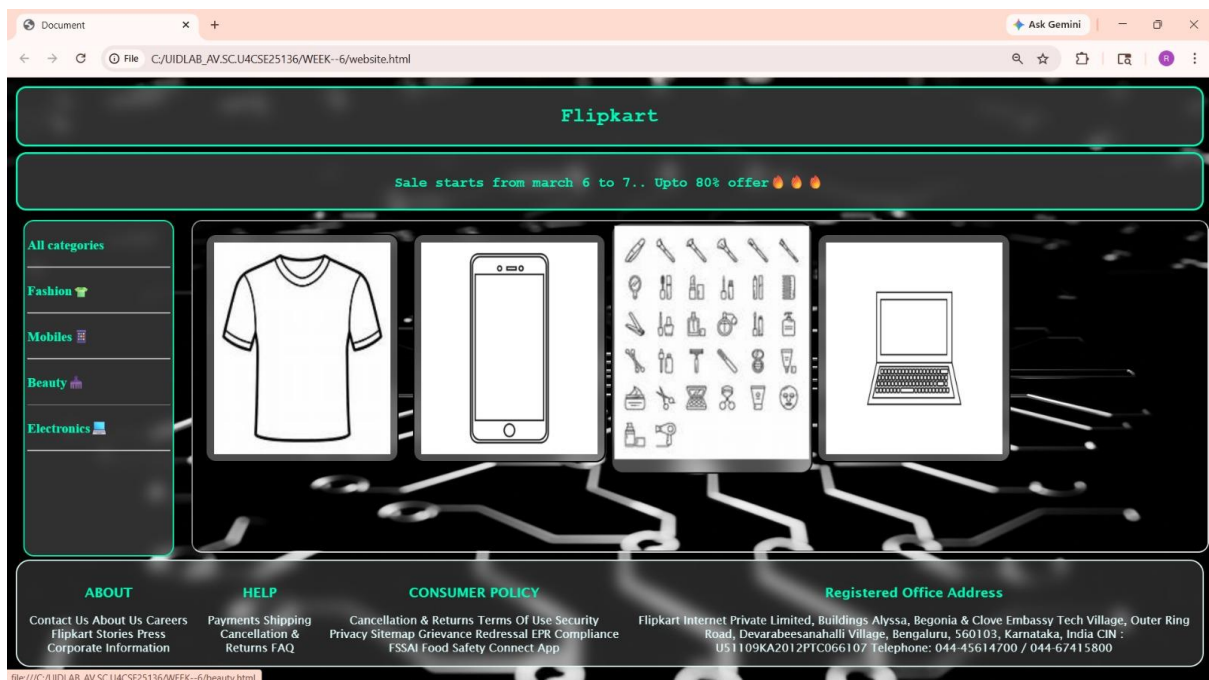
</div>

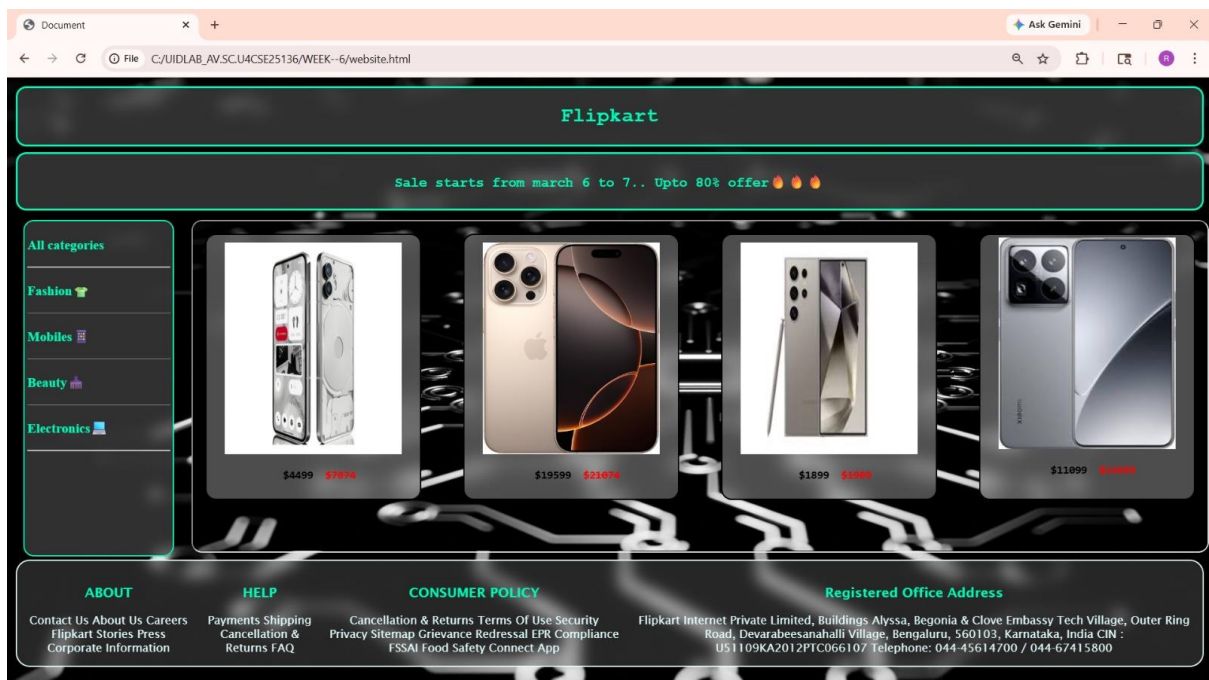
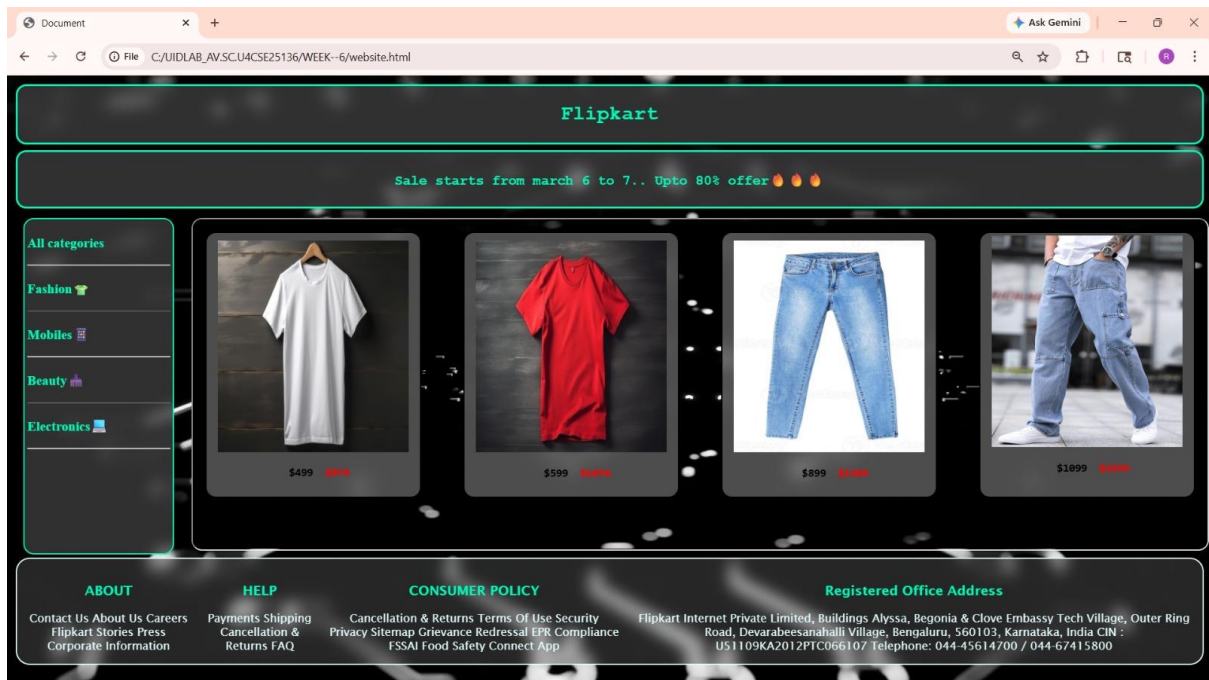
</body>

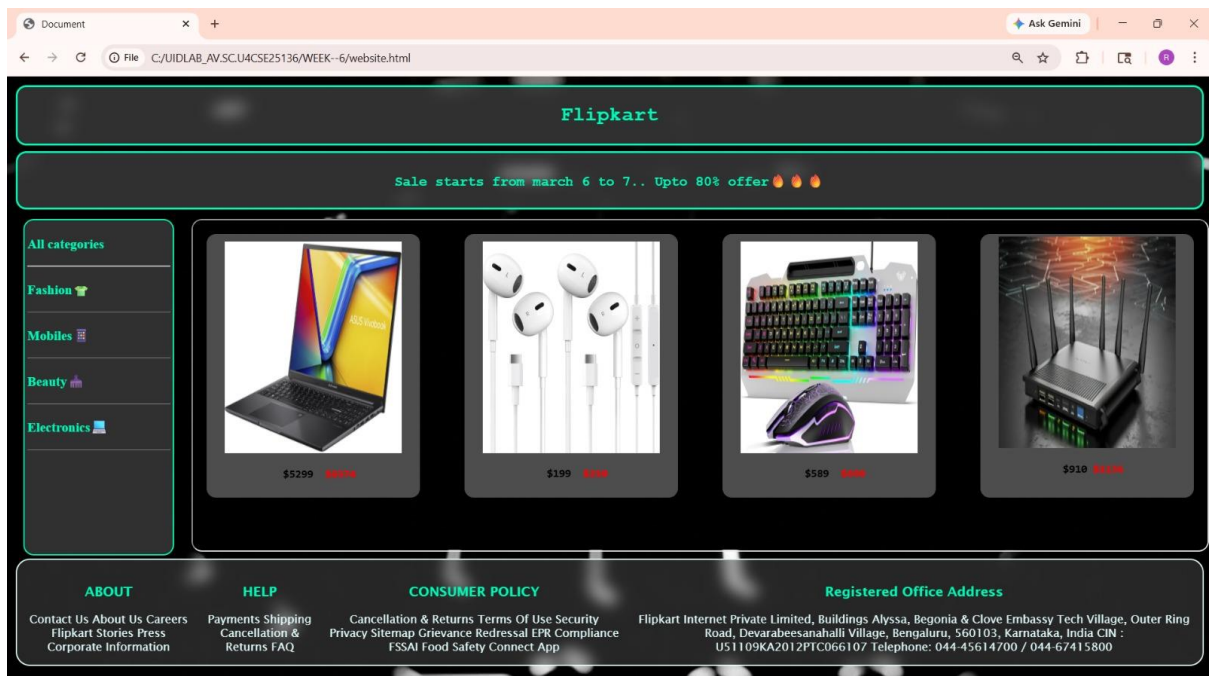
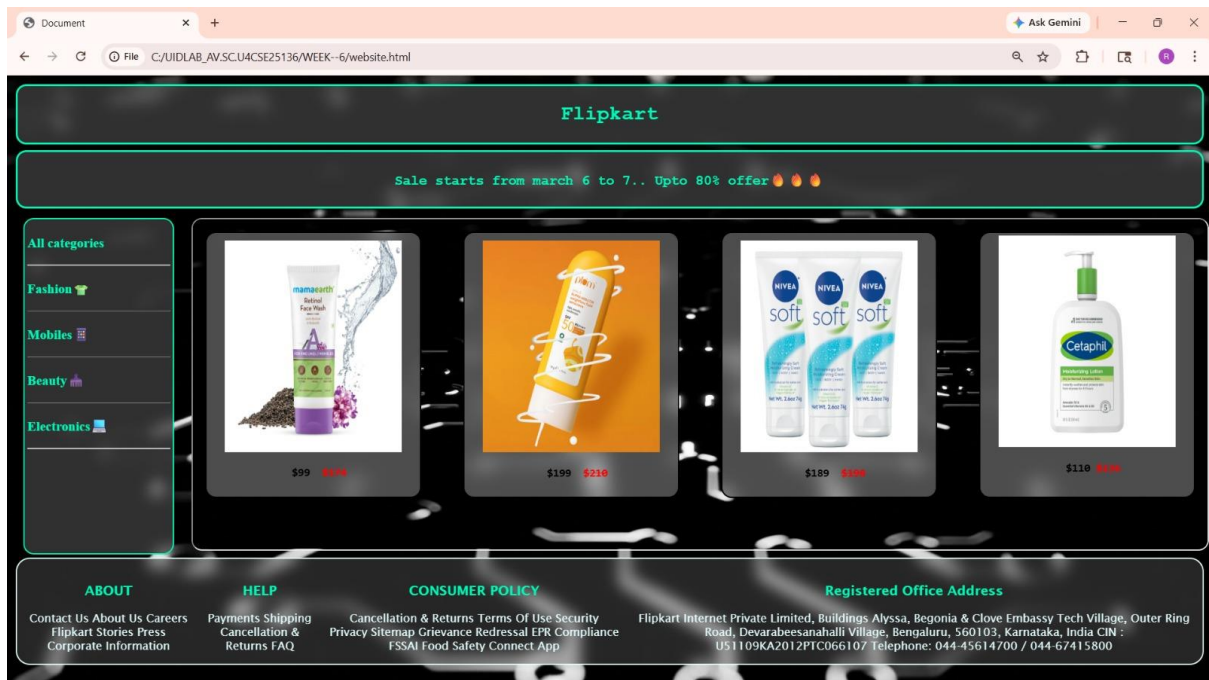
</html>

```

OUTPUT:







EXPLANATION:

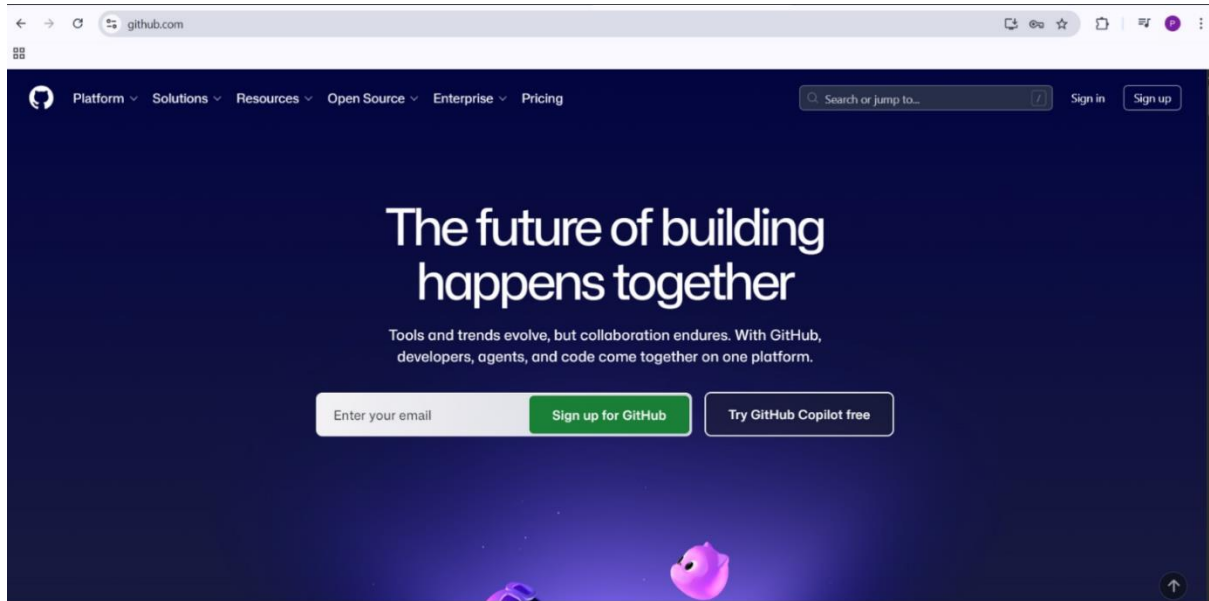
1. The .container class uses CSS Grid to arrange the webpage into different sections like header, alert, navigation, content, and footer.
2. grid-template-areas defines the layout structure of the grid with named areas such as head, alert, nav, site, and foot.
3. grid-template-columns: 15% 1fr creates two columns where the first column takes 15% width and the second takes the remaining space.
4. grid-template-rows sets the height of rows where the main content row occupies 70% of the height.
5. The .head class styles the header with a border, bold text, centered alignment, margin, and yellow background.
6. The .sale class styles the alert section with a border, bold centered text, margin, and orange background.
7. The .menu class styles the navigation area with a border, padding inside the box, and margins around it.
8. .allcategories iframe is used for the main content section and sets the iframe width to 100% and height to nearly full.
9. The .foot class creates the footer using flexbox with spacing between items, centered alignment, and a black background.
10. The .link class removes the underline from hyperlinks and changes their color to black.

Week-7

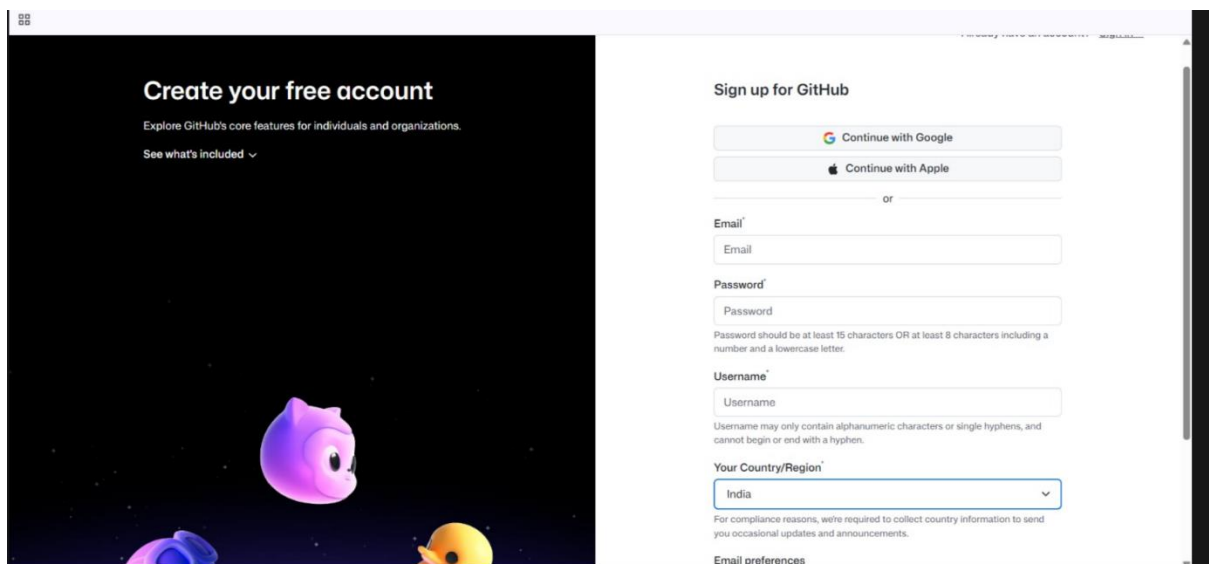
7.A) Aim: create a GitHub account and create a repository in it and push all the week 6 code files in it.

Procedure :

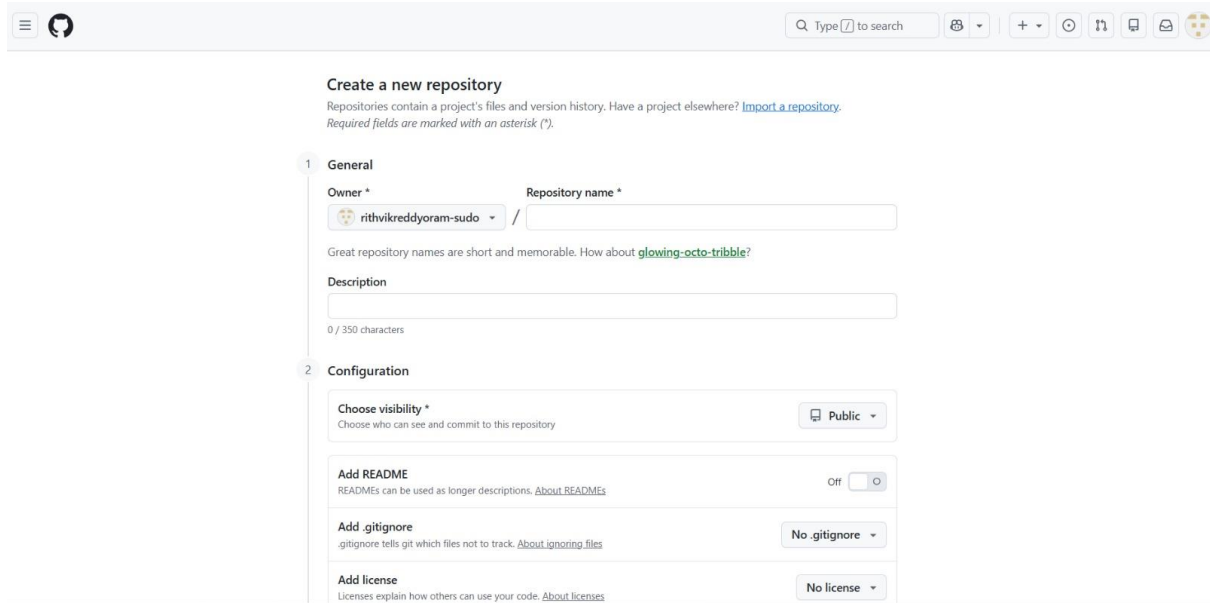
Step 1: browse for GitHub and enter it.



Step 2: login to Github and enter it.



Step3: Create a repository



The screenshot shows the 'Create a new repository' page on GitHub. At the top, there's a navigation bar with the GitHub logo, a search bar, and several utility icons. Below the navigation bar, the page title is 'Create a new repository'. A sub-header explains that repositories contain a project's files and version history, and provides a link to 'Import a repository'. A note states that required fields are marked with an asterisk (*).

The form is divided into two sections: '1 General' and '2 Configuration'.

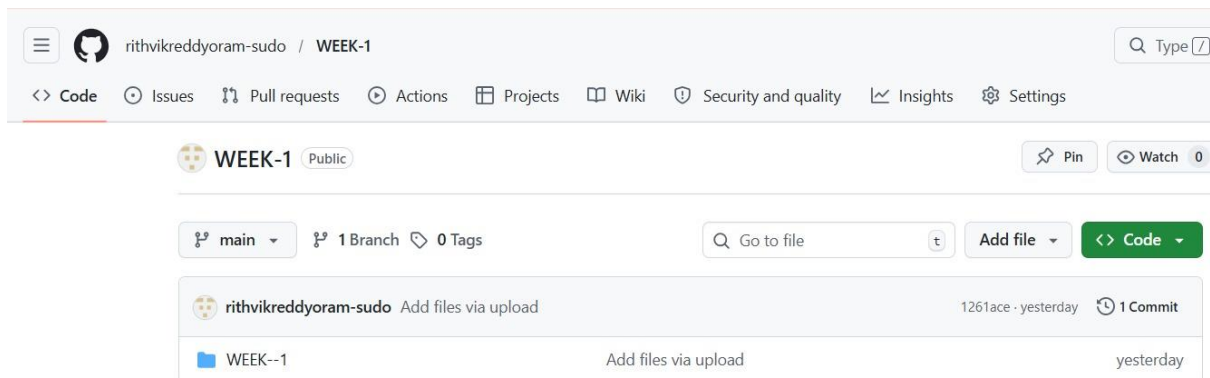
1 General

- Owner ***: A dropdown menu showing 'rithvikreddyoram-sudo'.
- Repository name ***: An empty text input field.
- A note: 'Great repository names are short and memorable. How about [glowing-octo-tribble?](#)'
- Description**: An empty text input field.
- A note: '0 / 350 characters'

2 Configuration

- Choose visibility ***: A dropdown menu showing 'Public'. Below it, a note says 'Choose who can see and commit to this repository'.
- Add README**: A section with a note 'READMEs can be used as longer descriptions. [About READMEs](#)'. There is a toggle switch set to 'Off'.
- Add .gitignore**: A section with a note '.gitignore tells git which files not to track. [About ignoring files](#)'. There is a dropdown menu showing 'No .gitignore'.
- Add license**: A section with a note 'Licenses explain how others can use your code. [About licenses](#)'. There is a dropdown menu showing 'No license'.

Step 4: Click on upload files and upload the files for the device storage.



The screenshot shows the GitHub repository page for 'WEEK-1' under the user 'rithvikreddyoram-sudo'. The repository is marked as 'Public'.

At the top, there's a navigation bar with the GitHub logo, the user name, the repository name 'WEEK-1', and a search bar. Below the navigation bar, there's a row of tabs: 'Code', 'Issues', 'Pull requests', 'Actions', 'Projects', 'Wiki', 'Security and quality', 'Insights', and 'Settings'.

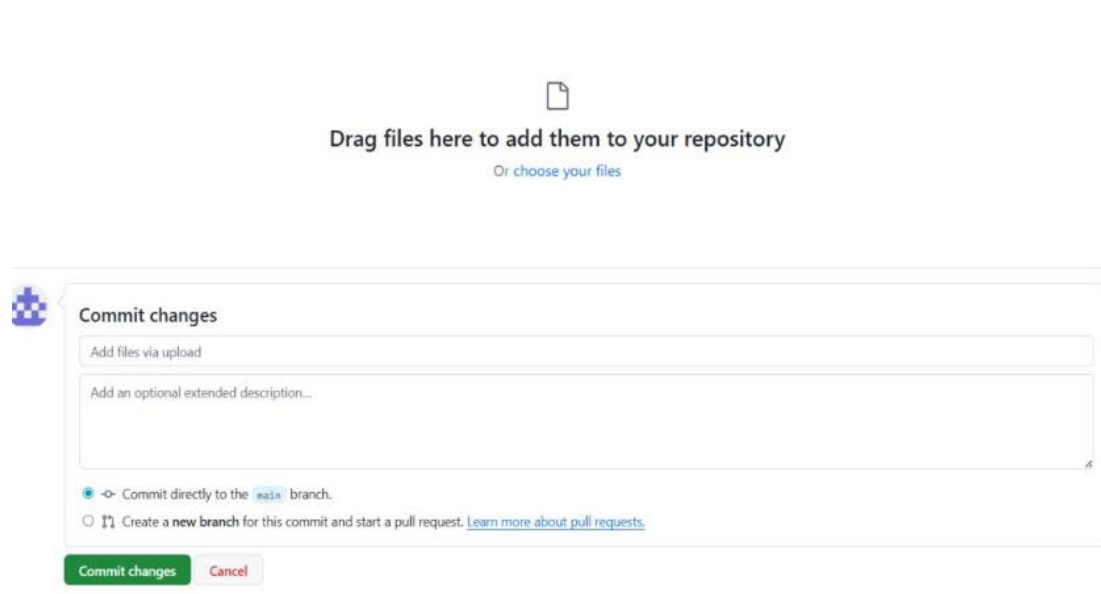
Below the tabs, there's a section for the repository 'WEEK-1' with a 'Public' badge. To the right, there are buttons for 'Pin' and 'Watch' (with a count of 0).

Below this, there's a section for the 'main' branch, showing '1 Branch' and '0 Tags'. There is a search bar 'Go to file' and buttons for 'Add file' and 'Code'.

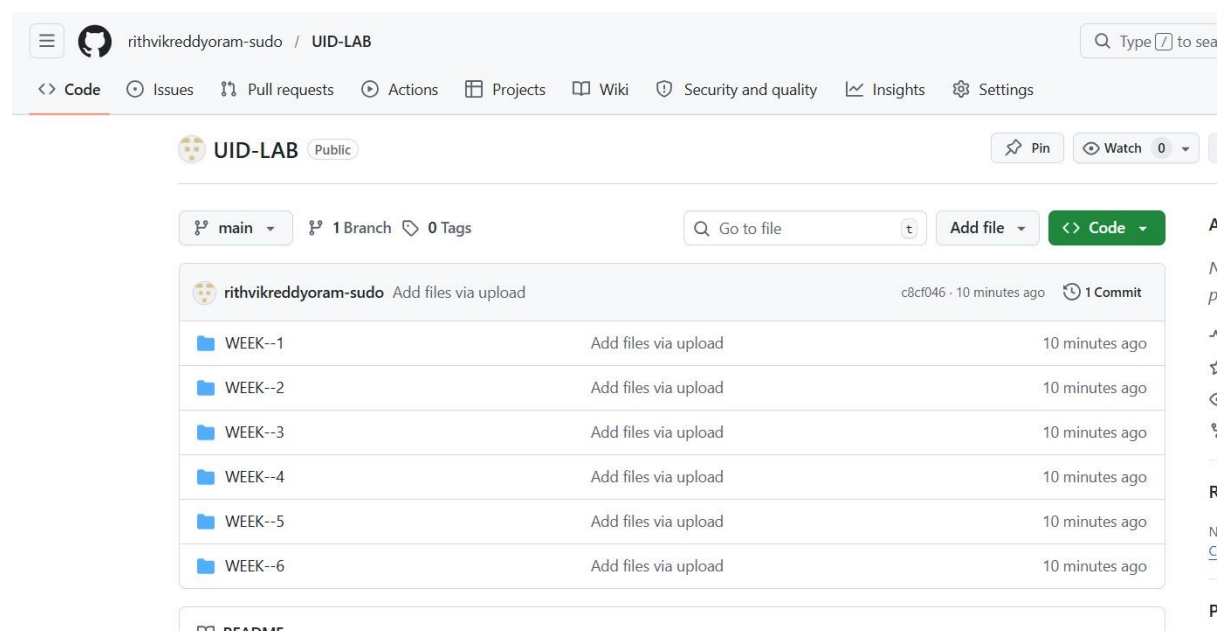
Below the branch section, there's a section for the repository 'rithvikreddyoram-sudo' with a note 'Add files via upload'. It shows the commit history: '1261ace · yesterday' and '1 Commit'.

Below the commit history, there's a section for the file 'WEEK--1' with a note 'Add files via upload' and the commit time 'yesterday'.

Step 5: Drag files into the given box and click on the commit changes.



Step 6:



Link for repository:

<https://github.com/rithvikreddyoram-sudo/UID-LAB>

WEEK-8

8A) Aim: To create a webpage that displays a square that continuously rotates 360° using css animations.

Program:

```
<!DOCTYPE html>

<html>

<head>

<title>Rotating Square</title>

<style>

  body {

    margin: 0;

    height: 100vh;

    display: flex;


    flex-direction: column;

    justify-content: flex-start;

    align-items: center;

    background-color: #2f3e46;

    font-family: Arial, sans-serif;

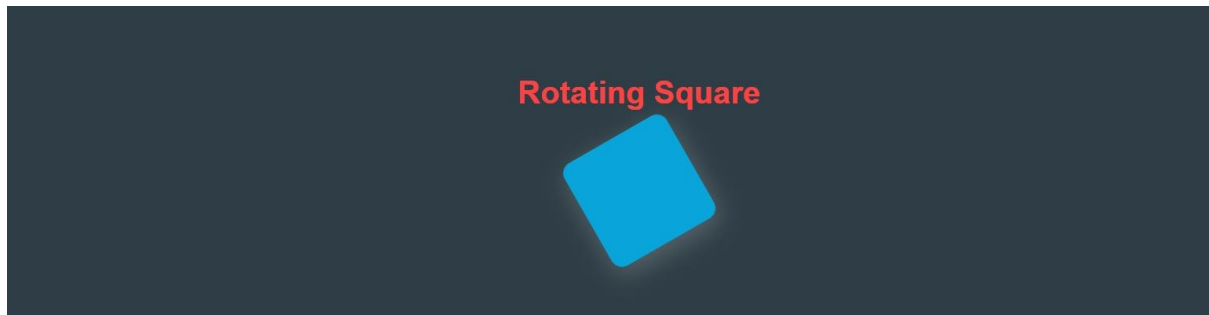
    padding-top: 50px;

  }

  .square {
```

```
width: 120px;
height: 120px;
background: #00bfffcc;
border-radius: 12px;
box-shadow: 0 8px 25px rgba(255, 255, 255, 0.2);
animation: rotate 3s linear infinite;
}
@keyframes rotate {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
</style>
</head>
<body>
  <h1 style="color: #ff4343;">Rotating Square</h1>
  <div class="square"></div>
</body>
</html>
```

OUTPUT:



EXPLANATION:

1. display: flex = It is used to easily align the items and distribute space within

the container.

2.flex-direction: column = is used to change the orientation of your layout from a horizontal row to a vertical column.

3.Align-items:centre = It is used to horizontally center the content because of the flex direction.

4.<div> = A <div> is an invisible container used to group HTML elements together so you can style them with CSS.

5.@keyframes = a keyframe tells the computer exactly how an element should look at a specific point in time.

6.Animation = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.

8B). AIM:

To create a ball that moves up and down continuously using css animations.

Program:

```
<!DOCTYPE html>

<html>

<head>

<title>Bouncing Ball</title>

<style>

body {

margin: 0;

height: 100vh;

display: flex;

flex-direction: column;

align-items: center;

justify-content: flex-start;

background: #b300ff;

}

h1 {

margin-top: 20px;

color: #ffea00;

}

.ball {
```

```
width: 80px;
height: 80px;
background: #00ddff;
border-radius: 50%;
margin-top: 120px;
animation: bounce 1.5s ease-in-out infinite;
}

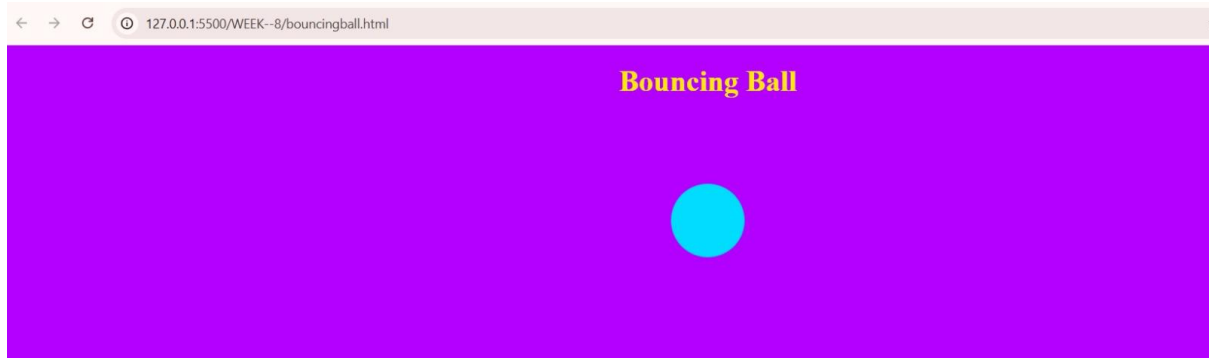
@keyframes bounce {
  0% {
    transform: translateY(0);
  }
  50% {
    transform: translateY(-120px);
  }
  100% {
    transform: translateY(0);
  }
}

</style>
</head>
<body>
  <h1>Bouncing Ball</h1>
  <div class="ball"></div>
```

```
</body>
```

```
</html>
```

OUTPUT:



Explanation:

1.flex-direction: column = is used to change the orientation of your layout from a horizontal row to a vertical column.

2.display: flex = It is used to easily align the items and distribute space within the container.

3.Align-items:centre = It is used to horizontally center the content because of the flex direction.

4.<div> = A <div> is an invisible container used to group HTML elements together so you can style them with CSS.

5.@keyframes = a keyframe tells the computer exactly how an element should look at a specific point in time.

6.Animation = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.

7.We denote percentages to tell at a particular time which work should be happen.

8.The transform: translateY(550px); property is a CSS tool used to move an element vertically along the Y-axis.

8C). AIM: To create a ball that moves horizontally and rotates at same time.

Program:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <title>Moving Rotating Circle</title>

  <style>

    body {

      margin: 0;

      height: 100vh;

      display: flex;

      flex-direction: column;

      align-items: center;

      justify-content: flex-start;

      background: #ff00d4;

      overflow: hidden;

    }

    h1 {

      margin-top: 20px;

      color: #00ffb7;

    }

    .circle {

      width: 80px;
```



```
height: 80px;
background: #00bbf5;
border-radius: 50%;
margin-top: 120px;
animation: moveRotate 3s linear infinite;
}
```

```
@keyframes moveRotate {
  0% {
    transform: translateX(-150px) rotate(0deg);
  }
  50% {
    transform: translateX(150px) rotate(180deg);
  }
  100% {
    transform: translateX(-150px) rotate(360deg);
  }
}
```

```
</style>
```

```
</head>
```

```
<body>
```

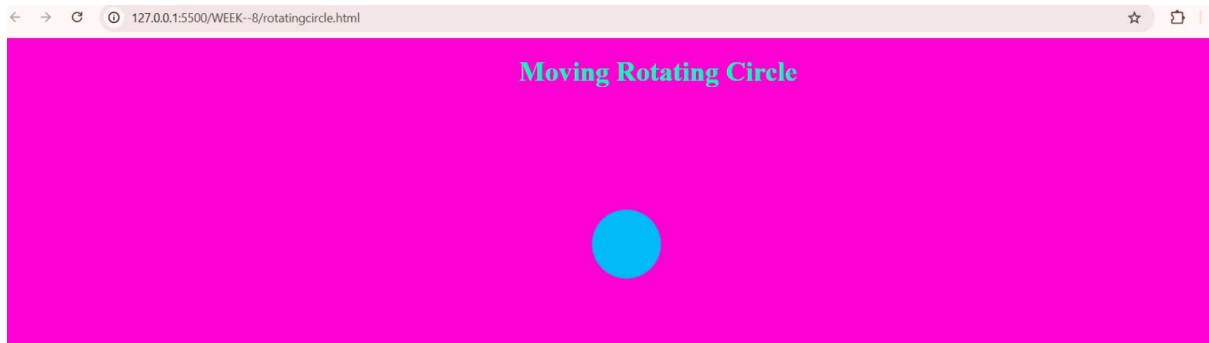
```
<h1>Moving Rotating Circle</h1>
```

```
<div class="circle"></div>
```

```
</body>
```

</html>

OUTPUT:



Explanation:

1.flex-direction: column = is used to change the orientation of your layout from a horizontal row to a vertical column.

2.display: flex = It is used to easily align the items and distribute space within the container.

3.Align-items:centre = It is used to horizontally center the content because of the flex direction.

4.<div> = A <div> is an invisible container used to group HTML elements together so you can style them with CSS.

5.@keyframes = a keyframe tells the computer exactly how an element should look at a specific point in time.

6.Animation = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.

7.We denote percentages to tell at a particular time which work should be happen.

8.The transform: translateX(__px); property is a CSS tool used to move an element vertically along the X-axis.

WEEK-9

9.A) Aim: : Week 9:Create a webpage using html,css,js based on the give conditions, create a button for each program , based on the button, clicked the corresponding program has to be executed and should display the logics for each button includes finding an even or odd number , prime or not , factorial of a number , Armstrong or not use appropriate design to display the output.

Program

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: linear-gradient(120deg, #2c3e50, #4ca1af);
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .container {
      background: #f5b8b8;
      padding: 30px;
      border-radius: 12px;
      width: 360px;
      text-align: center;
      box-shadow: 0 8px 20px rgba(0,0,0,0.3);
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>Logic Programs</h1>
  </div>
</body>
</html>
```

```
h2 {
  color: #ec3a3a;
  margin-bottom: 20px;
}

input {
  width: 100%;
  padding: 10px;
  border-radius: 8px;
  border: 1px solid #ccc;
  margin-bottom: 20px;
  text-align: center;
  font-size: 16px;
}

.buttons {
  display: flex;
  flex-direction: column;
  gap: 10px;
}

button {
  padding: 10px;
  border: none;
  border-radius: 8px;
  background: #14c904;
  color: white;
  font-weight: bold;
  cursor: pointer;
}

button:hover {
```

```

        background: #fff951;
    }

    .output {
        margin-top: 20px;
        padding: 15px;
        background: #f1f1f1;
        border-radius: 8px;
        min-height: 50px;
        color: #333;
    }
</style>
</head>

<body>

<div class="container">
    <h2>Logic Programs</h2>

    <input type="number" id="numInput" placeholder="Enter a number">

    <div class="buttons">
        <button onclick="checkEvenOdd()">Even / Odd</button>
        <button onclick="checkPrime()">Prime / Not</button>
        <button onclick="calcFactorial()">Factorial</button>
        <button onclick="checkArmstrong()">Armstrong</button>
    </div>

    <div class="output" id="output">Result</div>
</div>

<script>
function getInput() {

```

```

    const val = document.getElementById('numInput').value.trim();
    if (val === '' || isNaN(val)) {
        document.getElementById('output').textContent = 'Please
enter a valid number.';
        return null;
    }
    return parseInt(val);
}

function setOutput(text) {
    document.getElementById('output').textContent = text;
}

function checkEvenOdd() {
    const n = getInput();
    if (n === null) return;
    setOutput(n + ' is ' + (n % 2 === 0 ? 'Even' : 'Odd'));
}

function checkPrime() {
    const n = getInput();
    if (n === null) return;
    if (n < 2) { setOutput(n + ' is Not Prime'); return; }
    for (let i = 2; i <= Math.sqrt(n); i++) {
        if (n % i === 0) { setOutput(n + ' is Not Prime'); return; }
    }
    setOutput(n + ' is Prime');
}

function calcFactorial() {
    const n = getInput();
    if (n === null) return;

```

```

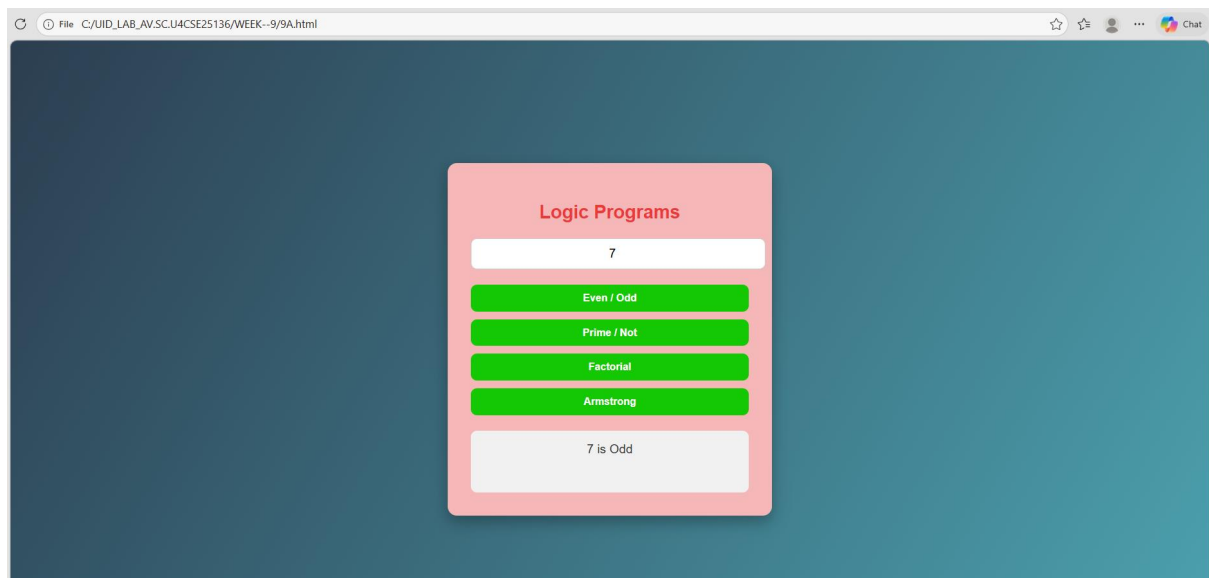
    if (n < 0) { setOutput('Factorial not defined for negative
numbers'); return; }
    let result = 1;
    for (let i = 2; i <= n; i++) result *= i;
    setOutput(n + '! = ' + result);
}

function checkArmstrong() {
    const n = getInput();
    if (n === null) return;
    const digits = String(n).split('');
    const power = digits.length;
    const sum = digits.reduce((acc, d) => acc +
Math.pow(Number(d), power), 0);
    setOutput(n + (sum === n ? ' is an Armstrong number' : ' is Not
an Armstrong number'));
}
</script>

</body>
</html>

```

OUTPUT:



Explanation:

DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this is an HTML document or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

<title> : the <title> element specifies a title for the HTML Page

<body> : the <body> element contains all visible content all visible content like text , image , links , tables , etc

<h> : we used <h> element for heading

<div>:we use <div> element for encapsulating the divisions and easy to modify

`<style>` : the `<style>` element is used to write CSS (designing part) it controls colors, layout, spacing, etc

`body` : the `body` selector is used to design the whole page here it sets background color and centers the content

`.container` : used to design the main box (card) gives white background, padding, shadow and rounded corners

`input` : used to design the input box width, padding, border and text alignment are set

`button` : used to design buttons gives color, size, border and hover effect

`.output` : used to design output box where result will be displayed

`<input>` : the `<input>` element is used to take user input `type="number"` allows only numbers `id` is used to access it in JavaScript

`<div class="buttons">` : groups all buttons together

`<button>` : used to perform actions when clicked `onclick` calls JavaScript functions

`<div id="output">` : used to display result initially shows "Result will appear here"

`<script>` : the `<script>` element is used to write JavaScript code it adds logic and functionality to the webpage

`getElementById()` : used to access HTML elements using `id`

`.value` : used to get value from input box

`.textContent` : used to display or change text inside an element

`function` : used to define a block of code that performs a task

`checkEvenOdd()` : checks whether number is even or odd uses $n \% 2$

`checkPrime()` : checks whether number is prime or not uses loop and division logic

`calcFactorial()` : calculates factorial of a number multiplies numbers from 1 to n

`checkArmstrong()` : checks Armstrong number sum of digits power equals number

`Math.sqrt()` : used to find square root (used in prime logic)

`parseInt()` : converts string to integer

`isNaN()` : checks if value is not a number

Week-10

10.A) Aim: write JavaScript conditional statement to find the sign of product of three numbers using functions .

Program:

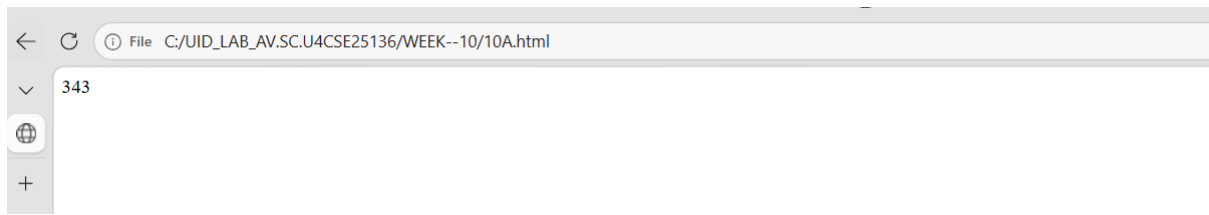
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Product of three numbers </title>
</head>
<body>
  <script>
    let a = parseInt(prompt("enter first number "));
    let b = parseInt(prompt("enter second number"));
    let c = parseInt(prompt("enter third number "));

    function product(a,b,c){
      return a*b*c;
    }

    document.write(product(a,b,c));

  </script>
</body>
</html>
```

OUTPUT:



Explanation:

The program determines the sign of the product of three numbers without focusing on the exact value.

1. The inputs are taken from three fields and converted into numeric values using `Number()`.
2. The product is calculated:

`let product = a * b * c;`

3. The sign of the result is determined using conditional checks:
 - If `product > 0` → the result is positive (+)
 - If `product < 0` → the result is negative (-)
 - If `product == 0` → the result is zero (0)
4. The output is displayed using `alert()`.
5. Logical Insight

Instead of multiplying, the sign could also be determined by:

- Counting negative numbers
- Even count → Positive
- Odd count → Negative

But in this implementation, direct multiplication is used for simplicity and clarity.

10.B)

Aim:

write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Sort Three Numbers</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #a36737,
#7d7d7d);
      height: 100vh;
      display: flex;
      justify-content: center;
      align-items: center;
      margin: 0;
    }

    .container {
      background: white;
      padding: 30px;
      border-radius: 15px;
      text-align: center;
      box-shadow: 0 10px 25px rgba(0,0,0,0.2);
      width: 300px;
```

```

    }

    input {
        width: 80%;
        padding: 10px;
        margin: 8px;
        border-radius: 8px;
        border: 1px solid #ccc;
    }

    button {
        margin-top: 15px;
        padding: 10px 20px;
        border: none;
        background: #b1b6bb;
        color: white;
        cursor: pointer;
    }
</style>
</head>

<body>

<div class="container">
    <h2>Sort 3 Numbers</h2>

    <input type="number" id="num1" placeholder="Enter
first number"><br>
    <input type="number" id="num2" placeholder="Enter
second number"><br>
    <input type="number" id="num3" placeholder="Enter
third number"><br>

```

```
<button onclick="sortNumbers()">Sort Numbers</button>
</div>

<script>
function sortNumbers() {

    let a = document.getElementById("num1").value;
    let b = document.getElementById("num2").value;
    let c = document.getElementById("num3").value;

    if (a === "" || b === "" || c === "") {
        alert("Please enter all numbers");
        return;
    }

    a = Number(a);
    b = Number(b);
    c = Number(c);

    let temp;

    if (a > b) {
        temp = a;
        a = b;
        b = temp;
    }

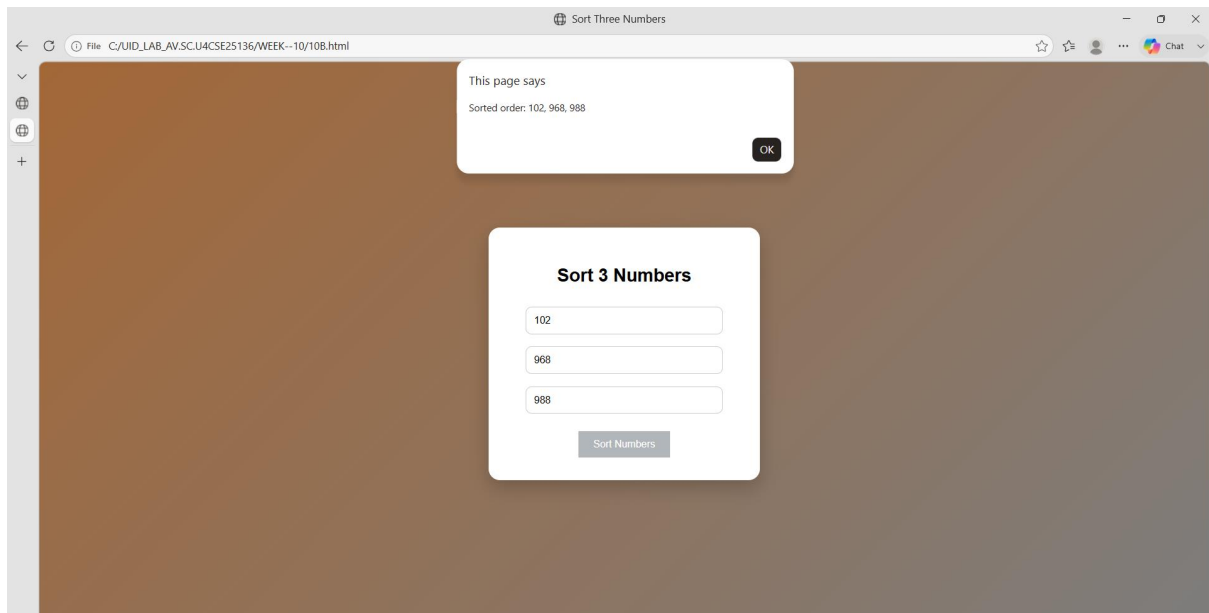
    if (a > c) {
        temp = a;
        a = c;
        c = temp;
    }
}
```

```
    if (b > c) {
        temp = b;
        b = c;
        c = temp;
    }

    alert("Sorted order: " + a + ", " + b + ", " + c);
}
</script>

</body>
</html>
```

Output:



Explanation:

Logic Explanation – *Sorting Three Numbers*

The program sorts three numbers in ascending order using comparison and swapping technique.

1. The input values are taken from three fields and converted into numeric values using Number().
2. A temporary variable temp is used for swapping values when needed.
3. The sorting is done using pairwise comparisons:

if (a > b) → swap a and b

if (a > c) → swap a and c

if (b > c) → swap b and c

4. After these three comparisons:
 - The smallest value will be stored in a
 - The middle value in b
 - The largest value in c
5. The sorted result is displayed using alert().

10.C)

Aim:

write a java script conditional statement to find largest of find largest of five numbers using function and onclick event .display an alert box to show the result

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Largest Number</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #242f91,
#02f9f5);
      height: 100vh;
      display: flex;
      justify-content: center;
      align-items: center;
      margin: 0;
    }

    .container {
      background: white;
      padding: 30px;
      border-radius: 15px;
      text-align: center;
      box-shadow: 0 10px 25px rgba(0,0,0,0.2);
```

```

        width: 300px;
    }

    input {
        width: 80%;
        padding: 10px;
        margin: 8px;
        border-radius: 8px;
        border: 1px solid #2ad18e;
    }

    button {
        margin-top: 15px;
        padding: 10px 20px;
        border: none;
        background: #ff0640;
        color: white;
        cursor: pointer;
    }
</style>
</head>

<body>

<div class="container">
    <h2>Largest of 5 Numbers</h2>
    <h3>Enter 5 numbers</h3>
    <h3>All numbers must be unique:</h3>
    <input type="number" id="num1" placeholder="Enter
first number"><br>
    <input type="number" id="num2" placeholder="Enter
second number"><br>

```

```

    <input type="number" id="num3" placeholder="Enter
third number"><br>
    <input type="number" id="num4" placeholder="Enter
fourth number"><br>
    <input type="number" id="num5" placeholder="Enter
fifth number"><br>

    <button onclick="findLargest()">Find Largest</button>
</div>

<script>
function findLargest() {

    let a = document.getElementById("num1").value;
    let b = document.getElementById("num2").value;
    let c = document.getElementById("num3").value;
    let d = document.getElementById("num4").value;
    let e = document.getElementById("num5").value;

    if (a === "" || b === "" || c === "" || d === "" || e
=== "") {
        alert("Please enter all numbers");
        return;
    }

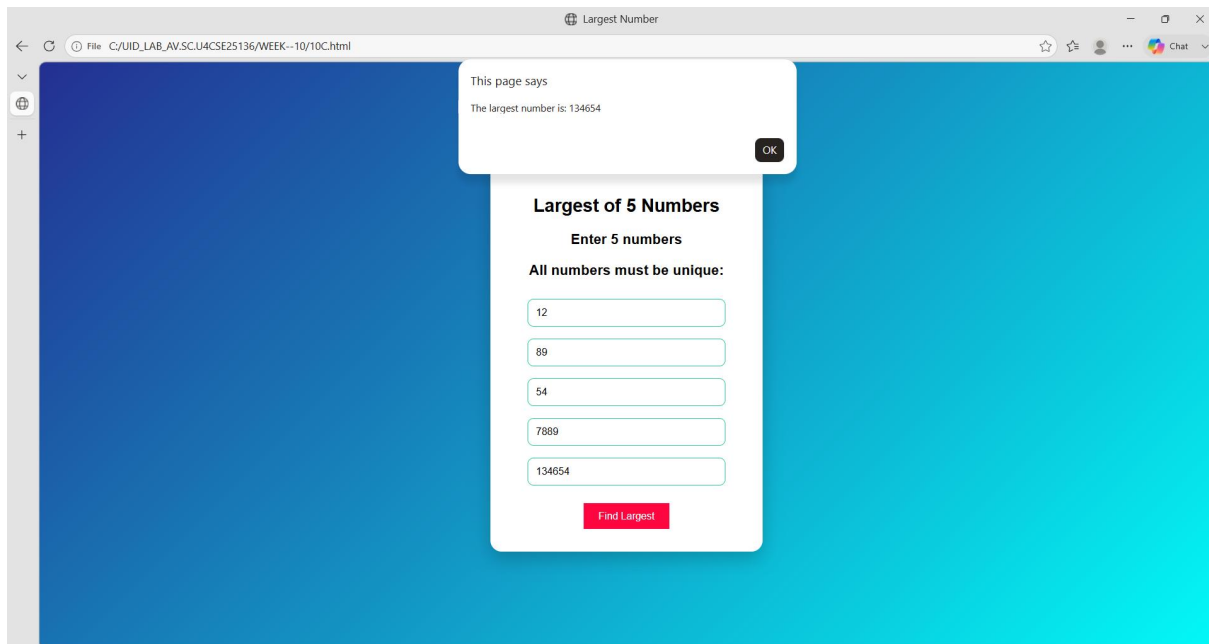
    a = Number(a);
    b = Number(b);
    c = Number(c);
    d = Number(d);
    e = Number(e);

    if (a===b || a===c || a===d || a===e ||
        b===c || b===d || b===e ||

```

```
        c===d || c===e ||  
        d===e) {  
            alert("Duplicate numbers not allowed!");  
            return;  
        }  
  
        let temp = a; // assume a is largest  
  
        if (b > temp) temp = b;  
        if (c > temp) temp = c;  
        if (d > temp) temp = d;  
        if (e > temp) temp = e;  
  
        alert("The largest number is: " + temp);  
    }  
</script>  
  
</body>  
</html>
```

Output:



Explanation:

Logic Explanation – *Largest of 5 Numbers*

The program finds the largest number among five unique inputs using sequential comparison logic.

1. The values are taken from five input fields and converted into numeric values using `Number()`.
2. Input validation ensures that all fields are filled. If any field is empty, execution stops.
3. A duplicate check is performed before processing:

`if (a===b || a===c || ... )`

- If any two numbers are equal → program stops

- Ensures all inputs are unique

4. The variable temp is initialized with the first value:

```
let temp = a;
```

5. Each remaining number is compared with temp:

```
if (b > temp) temp = b;
```

```
if (c > temp) temp = c;
```

```
if (d > temp) temp = d;
```

```
if (e > temp) temp = e;
```

6. After all comparisons:

- temp contains the largest number

7. The result is displayed using alert().

Logical Insight

Instead of using multiple complex conditions like:

```
if (a > b && a > c ...)
```

this approach uses incremental comparison, which is:

- More efficient
- Scalable
- Easier to maintain

10.D)

Aim:

write a java script for loop that will iterate from 0 to 15. For each iteration ,it will check if the current number odd or even , and display a message on screen

Program:

```
<!DOCTYPE html>

<html>
<head>
  <title>Even Odd Numbers</title>

<style>
  body {
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #2b5876, #d1ff05);
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    margin: 0;
  }

  .container {
    background: white;
    padding: 30px;
    border-radius: 15px;
    text-align: center;
```



```
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
    width: 350px;
}

input {
    width: 80%;
    padding: 10px;
    margin: 10px;
    border-radius: 8px;
    border: 1px solid #ccc;
}

button {
    padding: 10px 20px;
    border: none;
    background: #00ffbb;
    color: white;
    cursor: pointer;
    border-radius: 8px;
}

button:hover {
    background: #00c6ff;
}

#output {
    margin-top: 20px;
    text-align: left;
    background: #f5f5f5;
    padding: 10px;
    border-radius: 10px;
    max-height: 200px;
    overflow-y: auto;
}
```

```

        white-space: pre-line;
    }
</style>

</head>

<body>

<div class="container">
    <h2>Even & Odd Numbers</h2>

    <input type="number" id="numInput" placeholder="Enter
limit">
    <br>
    <button onclick="printNumbers()">Print Numbers</button>

    <div id="output"></div>

</div>

<script>
function printNumbers() {

    let n = document.getElementById("numInput").value;

    if (n === "") {
        alert("Enter limit");
        return;
    }

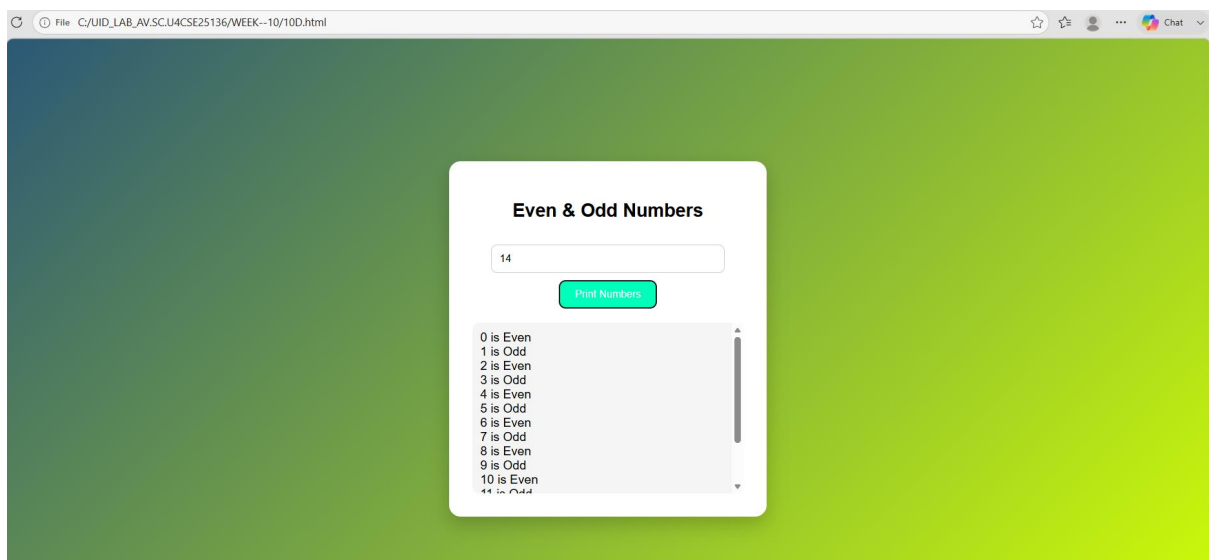
    n = Number(n);

    let result = "";

```

```
for (let i = 0; i <= n; i++) {  
    if (i % 2 === 0)  
        result += i + " is Even\n";  
    else  
        result += i + " is Odd\n";  
}  
  
document.getElementById("output").textContent =  
result;  
}  
</script>  
  
</body>  
</html>
```

Output:



Explanation:

Logic Explanation – *Even and Odd Numbers*

The program prints numbers from 0 to a given limit and identifies whether each number is even or odd.

1. The input value is taken from the input field using `document.getElementById()`.
2. Input validation ensures the field is not empty. If empty, execution stops.
3. The input is converted to a number using:

`n = Number(n);`

4. A loop is used to iterate from 0 to n:

`for (let i = 0; i <= n; i++)`

5. Each number is checked using the modulus operator:

`i % 2 === 0`

- If true → number is Even
- Else → number is Odd

6. The result is stored in a string and displayed inside the output div.

10.E)

Aim:

write a java script program to read 6 subjects marks from the student then computes the average marks of students then , this average is used to determine the corresponding grade.

Program:

```
<style>
  body {
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #1d976c,
#ff0051);
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    margin: 0;
  }

  .container {
    background: white;
    padding: 30px;
    border-radius: 15px;
    text-align: center;
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
    width: 320px;
  }

  input {
    width: 80%;
    padding: 10px;
    margin: 6px;
```

```

        border-radius: 8px;
        border: 1px solid #ccc;
    }

    button {
        margin-top: 15px;
        padding: 10px 20px;
        border: none;
        background: #dc1212;
        color: white;
        cursor: pointer;
        border-radius: 8px;
    }

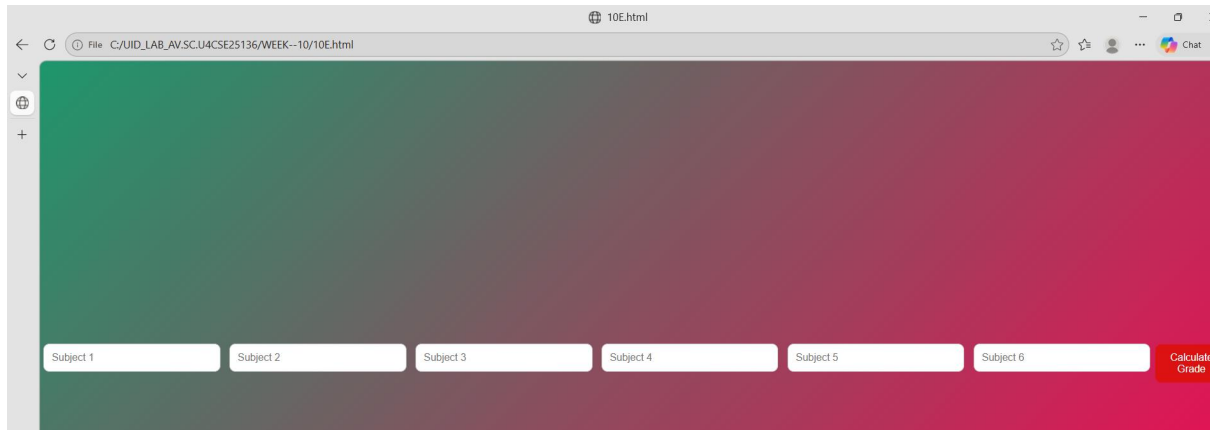
    #output {
        margin-top: 15px;
        font-weight: bold;
    }
</style>
<input type="number" id="m1" placeholder="Subject 1"><br>
<input type="number" id="m2" placeholder="Subject 2"><br>
<input type="number" id="m3" placeholder="Subject 3"><br>
<input type="number" id="m4" placeholder="Subject 4"><br>
<input type="number" id="m5" placeholder="Subject 5"><br>
<input type="number" id="m6" placeholder="Subject 6"><br>

<button onclick="calculateGrade()">Calculate
Grade</button>

<div id="output"></div>

```

Output:



Explanation:

Logic Explanation – *Student Grade Calculation*

The program calculates the average marks of 6 subjects and assigns a grade based on conditions.

1. Six subject marks are taken as input from the user.
2. Input validation ensures all fields are filled.
3. Values are converted into numeric type using `Number()`.
4. The total sum is calculated:

```
sum = a + b + c + d + e + f;
```

5. The average is computed:

```
avg = sum / 6;
```

6. Grade is assigned using conditional logic:

```
if (avg >= 90) grade = "A";  
else if (avg >= 75) grade = "B";  
else if (avg >= 60) grade = "C";  
else grade = "D";
```

The result (average + grade) is displayed on the screen

WEEK-11

11A) Aim: A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

Program :

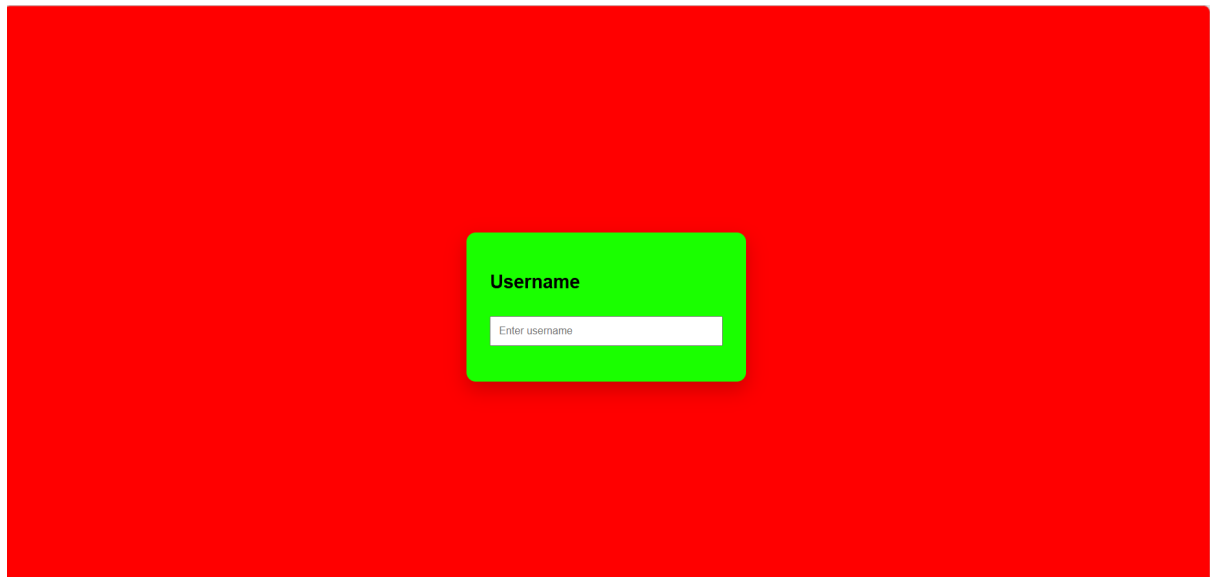
```
<html>
<head>
<title>Username Check</title>
<style>
body {
  margin:0;
  font-family: Arial;
  background: linear-gradient(120deg,#ff0000,#ff0000);
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}
.card {
  background:rgb(26, 255, 0);
  padding:30px;
  border-radius:12px;
  width:300px;
  box-shadow:0 10px 25px rgba(0,0,0,0.2);
}
input { width:100%; padding:10px; margin-top:10px; }
</style>
</head>
<body>
```



```
<div class="card">
<h2>Username</h2>
<input type="text" id="username" placeholder="Enter
username"
onkeyup="checkUsername()">
<p id="msg"></p>
</div>

<script>
function checkUsername() {
  let val=document.getElementById("username").value;
  let msg=document.getElementById("msg");
  if(val.length<5){ msg.innerText="Minimum 5 characters
required";
msg.style.color="red"; }
  else{ msg.innerText="Valid username";
msg.style.color="green"; }
}
</script>
</body>
</html>
```

OUTPUT:



Explantion:

<!DOCTYPE html> : the <!DOCTYPE html> element tells the browser that this is an HTML5 document

<html> : the <html> element is the root element of the web page all elements are written inside this

<head> : the <head> element contains meta information like title and styles

<title> : the <title> element specifies the name of the web page it appears on the browser tab

<style> : the <style> element is used to write CSS for designing the page

body : the body selector is used to style the whole page here flex is used to center the card layout

.card : this class is used to create a box container with padding shadow and rounded corners

input : the <input> element is used to take user input for username

<p> : the <p> element is used to display validation message

`<script>` : the `<script>` element is used to write JavaScript code

`function` : used to define a block of code that performs username validation

`getElementById()` : used to access username input and message element

`.value` : used to get value from input box

`length` : used to check number of characters entered

`if condition` : used to check if username is less than 5 characters

`innerText` : used to display message on screen

`style.color` : used to change text color based on validation

11B) Aim: A registration form should validate email format before submission. If invalid, the form should not be submitted

Program:

```
<!DOCTYPE html>
<html>
<head>
<title>Email Validation</title>

<style>
body {
    margin: 0;
    font-family: Arial;
    background: linear-gradient(120deg, #ff6a00, #ee0979);
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

.card {
    background: rgb(8, 232, 37);
    padding: 30px;
    border-radius: 12px;
    width: 320px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

input, button {
    width: 100%;
    padding: 10px;
```

```

    margin-top: 10px;
}

button {
    background: #ee0979;
    color: white;
    border: none;
    cursor: pointer;
}

button:hover {
    background: #c80060;
}
</style>

</head>

<body>

<div class="card">
    <h2>Email Check</h2>

    <form onsubmit="return validateEmail()">
        <input type="text" id="email" placeholder="Enter Email">
        <button type="submit">Submit</button>
    </form>
</div>

<script>
function validateEmail() {
    let email = document.getElementById("email").value;
    let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

```

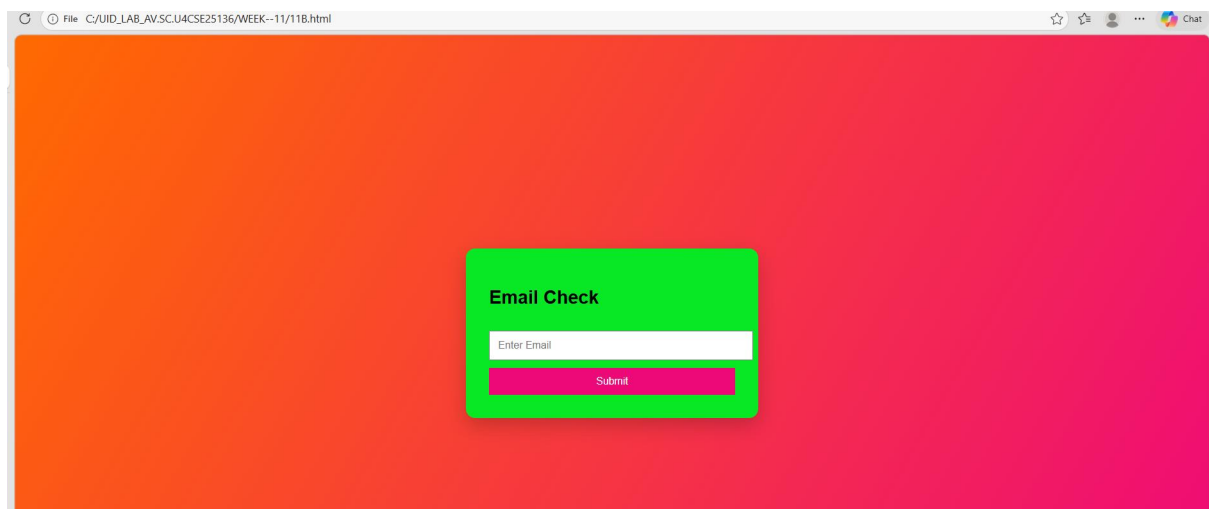
```

    if (pattern.test(email)) {
        alert("Email is valid");
        return true;
    } else {
        alert("Invalid Email");
        return false;
    }
}
</script>

</body>
</html>

```

OUTPUT:



Explantion:

<!DOCTYPE html> : the <!DOCTYPE html> element tells the browser that this is an HTML5 document

<html> : the <html> element is the root element of the web page

<head> : contains title and styles

<title> : displays page name on browser tab

<style> : used to design the page with gradient background and card layout

form : the <form> element is used to collect user input

input : used to take email input from user

button : used to submit the form

onsubmit : used to call JavaScript function when form is submitted

<script> : used to write JavaScript validation logic

function : defines email validation logic

getElementById() : used to access email input

.value : gets the email value entered by user

pattern : regular expression used to check valid email format

test() : checks if email matches pattern

if condition : checks whether email is valid or not

alert() : used to display popup message for valid or invalid email

return false : prevents form submission if invalid

return true : allows form submission if valid

11.C) Aim: A system should check password strength while typing and display whether it is weak or strong.

Program :

```
<!DOCTYPE html>
<html>
<head>
<title>Password Strength</title>

<style>
body {
```

```
margin: 0;
font-family: Arial;
background: linear-gradient(120deg, #ff0613, #0199ff);
display: flex;
justify-content: center;
align-items: center;
height: 100vh;
}

.card {
  background: rgb(146, 255, 3);
  padding: 30px;
  border-radius: 12px;
  width: 350px;
  box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

input {
  width: 100%;
  padding: 10px;
  margin-top: 10px;
  border: 2px solid #4f46ff;
  border-radius: 6px;
}

#msg {
  font-weight: bold;
  margin-top: 10px;
}
</style>

</head>
```



```

<body>

<div class="card">
  <h2>Password Strength</h2>

  <p><b>Rules:</b><br>
    • At least 6 characters<br>
    • One uppercase letter<br>
    • One special symbol</p>

  <input type="password" id="pass"
onkeyup="checkStrength()">
  <p id="msg"></p>
</div>

<script>
function checkStrength() {
  let pass = document.getElementById("pass").value;
  let msg = document.getElementById("msg");

  let strong = /^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;
  let medium = /^(?=.*[A-Z]).{6,}$/;

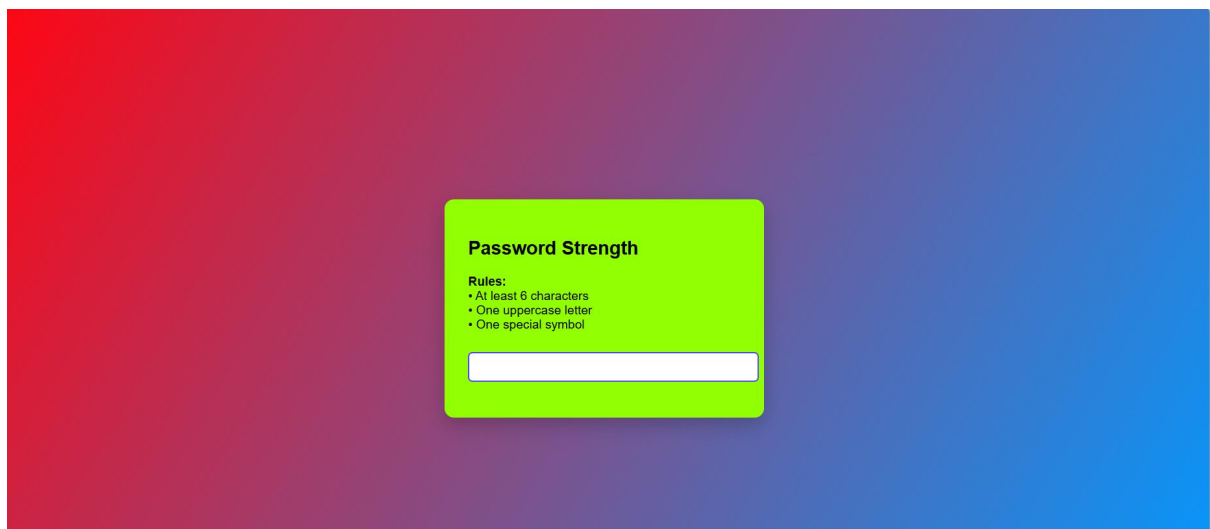
  if (strong.test(pass)) {
    msg.innerText = "Strong Password";
    msg.style.color = "green";
  }
  else if (medium.test(pass)) {
    msg.innerText = "Medium Password";
    msg.style.color = "orange";
  }
  else {
    msg.innerText = "Weak Password";
  }
}

```

```
        msg.style.color = "red";
    }
}
</script>

</body>
</html>
```

OUTPUT:



Explantion:

<!DOCTYPE html> : defines HTML5 document

<html> : root element of webpage

<head> : contains title and style

<title> : shows page title

<style> : used for page design and layout styling

body : used to center the card with gradient background

.card : used to create container box

<p> : used to display password rules

input : used to enter password

<script> : contains JavaScript logic

function : defines password checking logic

getElementById() : used to access password input and message

.value : gets password entered by user

pattern : regular expression used to check password strength

(?=.*[A-Z]) : ensures at least one uppercase letter

(?=.*[!@#\$%^&*]) : ensures at least one special symbol

.{6,} : ensures minimum 6 characters

test() : checks if password matches pattern

if condition : checks strong or weak password

innerText : displays result message

style.color : changes text color based on strength

11.D) Aim : A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits

Program :

```
<!DOCTYPE html>
<html>
<head>
<title>Mobile Validation</title>

<style>
body {
    margin: 0;
    font-family: Arial;
    background: linear-gradient(120deg, #19ff38, #00ffae); /*
changed color */
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

.card {
    background: rgb(234, 255, 0);
    padding: 30px;
    border-radius: 12px;
    width: 320px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

input {
```

```

        width: 100%;
        padding: 10px;
        margin-top: 10px;
        border-radius: 6px;
        border: 2px solid #ccc;
    }

    #msg {
        margin-top: 10px;
        font-weight: bold;
    }
</style>

</head>

<body>

<div class="card">
    <h2>Mobile & Password</h2>

    <input type="text" id="mobile" placeholder="Enter Mobile"
onblur="validateMobile()">
    <input type="password" id="pass" placeholder="Enter Password"
onkeyup="checkPass()">

    <p id="msg"></p>
</div>

<script>
function validateMobile() {
    let num = document.getElementById("mobile").value;
    let msg = document.getElementById("msg");
    let pattern = /^[0-9]{10}$/;

```

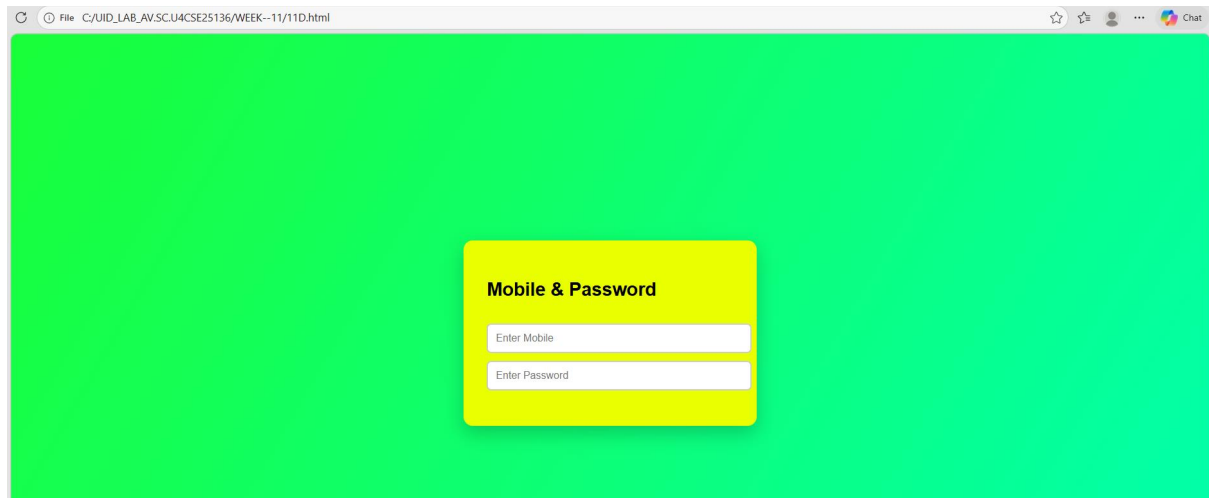
```
    if (pattern.test(num)) {
        msg.innerText = "Valid Mobile Number";
        msg.style.color = "green";
    } else {
        msg.innerText = "Invalid Mobile Number";
        msg.style.color = "red";
    }
}

function checkPass() {
    let pass = document.getElementById("pass").value;
    let strong = /^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;
    let msg = document.getElementById("msg");

    if (strong.test(pass)) {
        msg.innerText = "Strong Password";
        msg.style.color = "green";
    } else {
        msg.innerText = "Weak Password";
        msg.style.color = "red";
    }
}
</script>

</body>
</html>
```

OUTPUT:



Explantion:

<!DOCTYPE html> : defines HTML5 document

<html> : root element of page

<head> : contains title and styles

<title> : sets page title

<style> : used for UI design and layout

body : centers the content using flexbox

.card : used as container box

input : used to take mobile number and password input

onblur : triggers validation when user leaves mobile input field

onkeyup : triggers password validation while typing

<script> : contains JavaScript code

function : defines validation logic

getElementById() : used to access input fields and message

.value : gets input values

pattern : regular expression for mobile validation

^[0-9]{10}\$: ensures exactly 10 digits

test() : checks if mobile number is valid

if condition : checks valid or invalid mobile

password pattern : same as strong password rule

innerText : displays result message

11.E) Aim: 5. Design an HTML page, such that the registration form must validate: Name (not empty) Email (valid format) Password (minimum 6 characters)

Program:

```
<!DOCTYPE html>
<html>
<head>
<title>Register</title>

<style>
body{
    margin:0;
    font-family:Arial;
    background: linear-gradient(120deg, #0073ff, #0077ff);
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
    color:white;
}

.card{
    background:rgb(255, 252, 155);
    color:black;
    padding:30px;
    border-radius:12px;
    width:350px;
    box-shadow:0 10px 25px rgba(0,0,0,0.3);
}
```

```

input,button{
    width:100%;
    padding:10px;
    margin-top:10px;
}

button{
    background:#243b55;
    color:white;
    border:none;
    cursor:pointer;
}

button:hover{
    background:#141e30;
}
</style>

</head>

<body>

<div class="card">
<h2>Registration Form</h2>

<form onsubmit="return validateForm()">
<input type="text" id="name" placeholder="Name">
<input type="text" id="email" placeholder="Email">
<input type="password" id="pass" placeholder="Password">
<button type="submit">Register</button>
<p id="msg"></p>
</form>

```

```
</div>

<script>
function validateForm(){
    let name=document.getElementById("name").value;
    let email=document.getElementById("email").value;
    let pass=document.getElementById("pass").value;
    let msg=document.getElementById("msg");

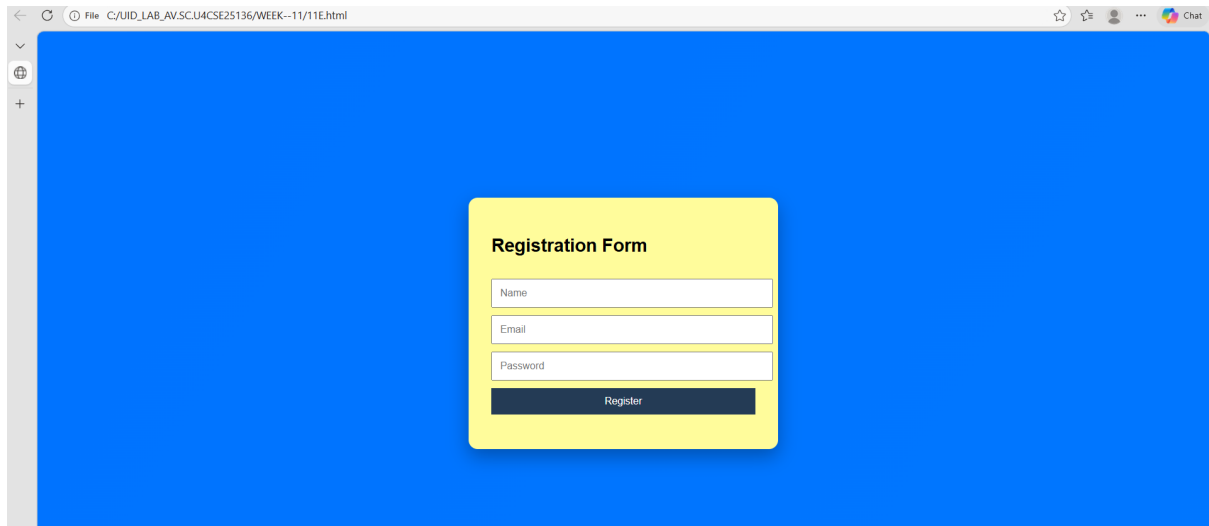
    let emailPattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    let strong=/^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;

    if(name===""){ msg.innerText="Name required"; return false; }
    if(!emailPattern.test(email)){ msg.innerText="Invalid email"; return
false; }
    if(!strong.test(pass)){ msg.innerText="Weak password"; return false; }

    alert("Registration Successful");
    return true;
}
</script>

</body>
</html>
```

OUTPUT:



Explantion:

<!DOCTYPE html> : defines HTML5 document

<html> : root element of webpage

<head> : contains title and style

<title> : sets name of webpage

<style> : used to design form layout and appearance

body : centers form with gradient background

.card : used to create form container

form : used to collect user data

input : used to take name email and password

button : used to submit form

onsubmit : triggers validation before submission

<script> : contains JavaScript validation logic

function : defines form validation process

getElementById() : used to access form fields

.value : gets input values

emailPattern : checks valid email format

strong pattern : checks password strength

if condition : checks name not empty

if condition : checks email format

if condition : checks password rules

alert() : shows success message when all validations pass

return false : prevents submission if invalid

return true : allows submission if valid

innerText : displays error messages

WEEK-12

12(a) Aim: To install and setup the Electron Framework application.

Procedure:

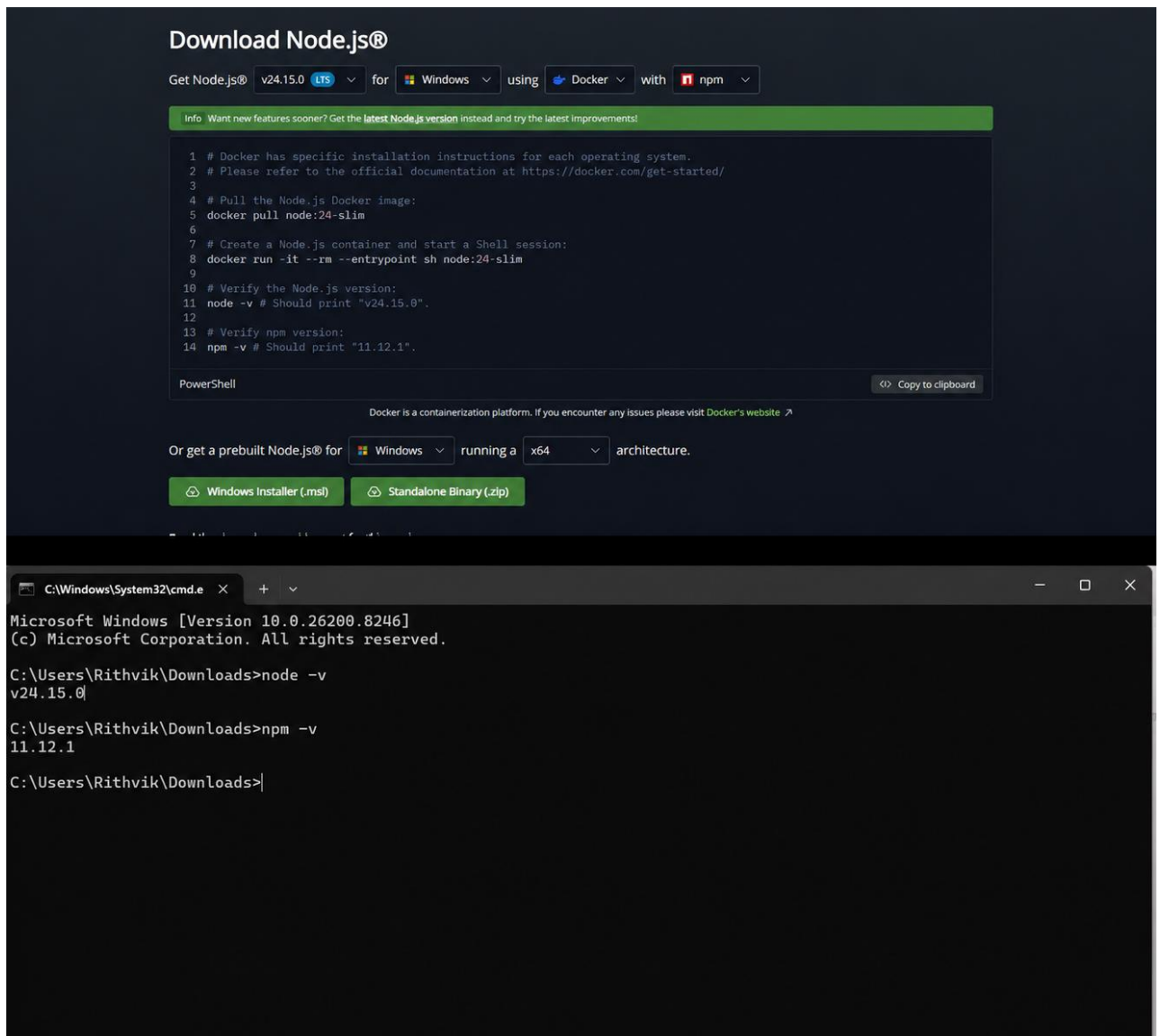
Step 1: Go to the web browser and download node.js.

Step 2: After installing the Node.js, complete the setup process.

Step 3: After installing node.js, Open the command prompt and navigate to the folder where your project is located.

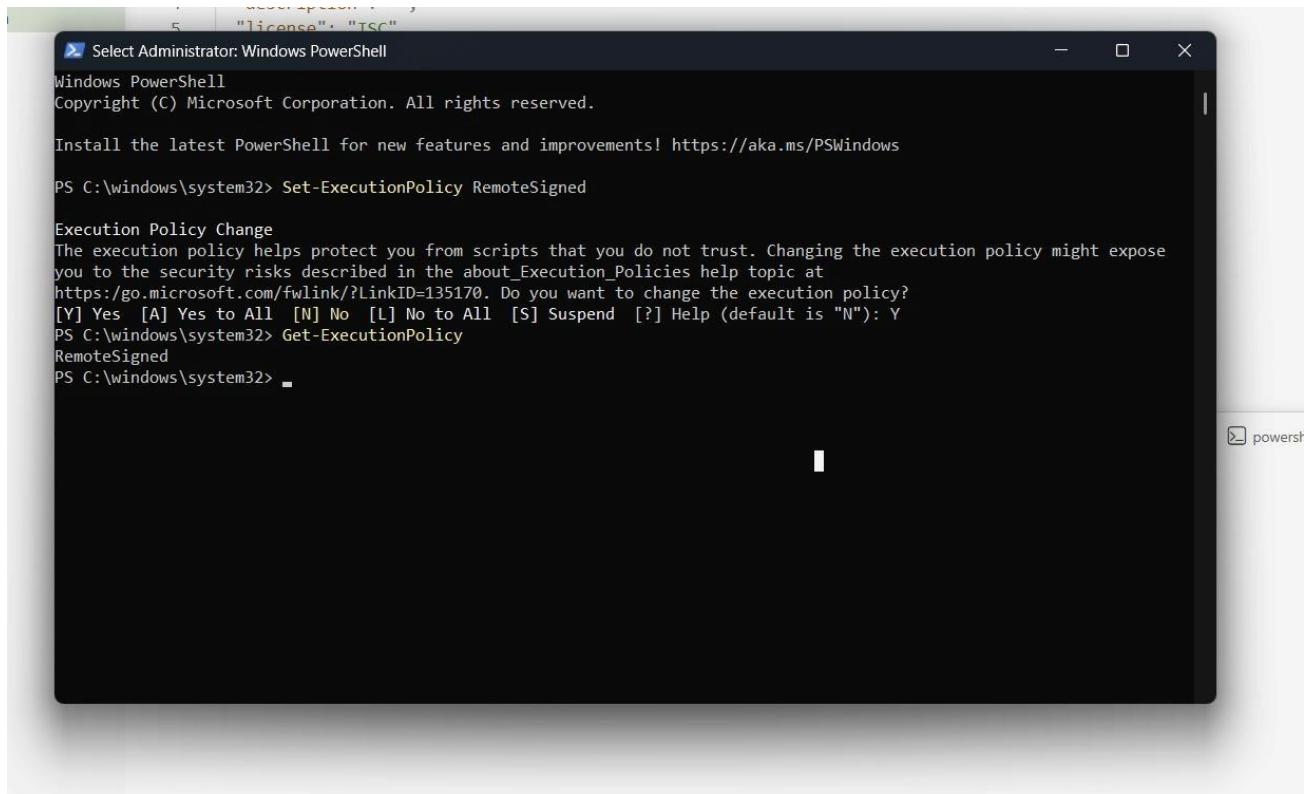
Step 4: And give the prompt “node –v” and also "npm –v".

Step 5: If you get the versions then it is successfully installed.

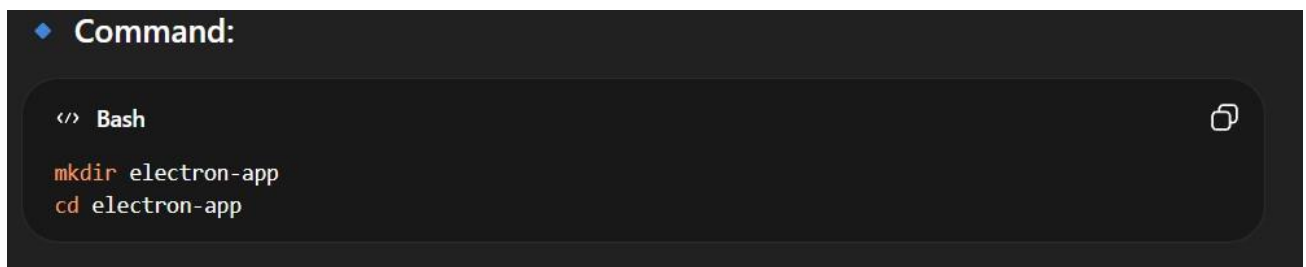


Step 6: Now open PowerShell as Administrator and give the prompt "Set-ExecutionPolicy RemoteSigned". 138

- This command allows local PowerShell scripts to run securely.
- And press Y.



Create Project Folder :



Step 7: Create a new folder for the Electron project using mkdir.

Step 8: Move into the project folder using cd.

This helps keep all project files organized.

Step 9:

Initialize Node Project:

```
◆ Command:

</> Bash
npm init -y
```

The above command creates the package.json file.

This file stores:

1. Project name
2. Version
3. Main entry file
4. Scripts
5. Dependencies

Without package.json file, npm cannot manage project properly.

Step 10:

Install Electron:

```
◆ Command:

</> Bash
npm install electron --save-dev
```

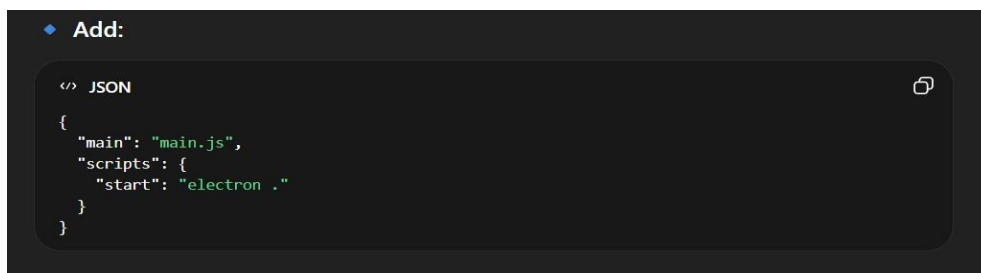
1. Downloads Electron package
2. Saves it in node_modules

3. Adds it to devDependencies

4.--save-dev→Because Electron is needed during development.

Step 11:

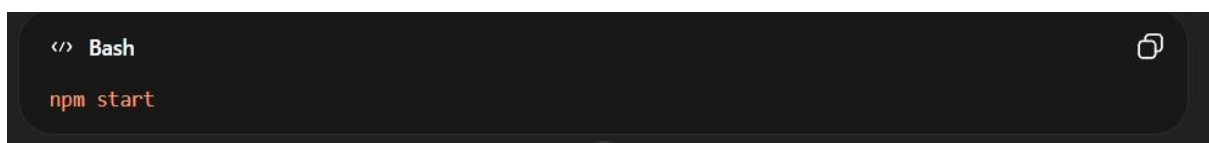
Configure package.json:



```
◆ Add:
</> JSON
{
  "main": "main.js",
  "scripts": {
    "start": "electron ."
  }
}
```

Main→Tells Electron which file starts the application.

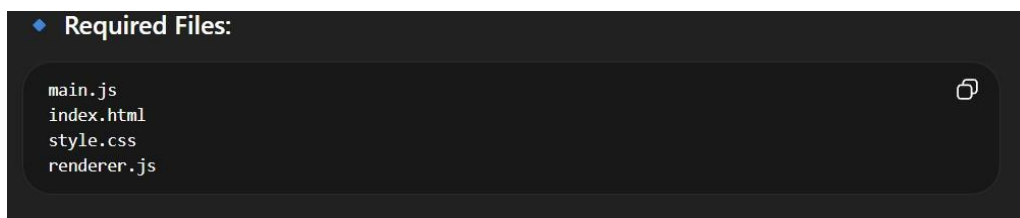
Start→Defines command used when we run:



```
</> Bash
npm start
```

Electron → This command runs Electron in the current project folder.

Create Main Application Files:



```
◆ Required Files:
main.js
index.html
style.css
renderer.js
```

Step 12: main.js→Controls app lifecycle and creates windows

Step 13: index.html→User interface

Step 14: style.css→Design and layout

Step 15: renderer.js→Handles user interaction

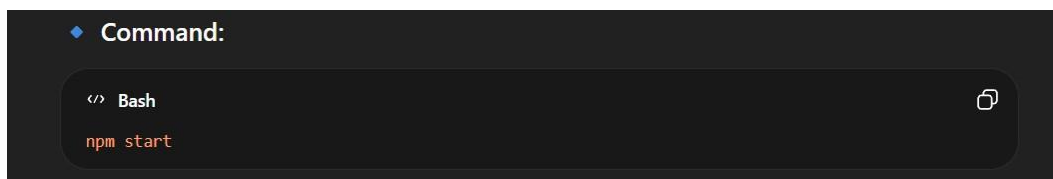
Write main.js

- 1.Imports Electron modules
- 2.Creates application window
- 3.Loads UI



```
1  const { app, BrowserWindow } = require('electron');
2
3  function createWindow() {
4    const win = new BrowserWindow({
5      width: 800,
6      height: 600
7    });
8
9    win.loadFile('index.html');
10 }
11
12 app.whenReady().then(createWindow);
```

Run Application:



```
Command:
</> Bash
npm start
```